X-Plane MD302

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The pop-up MD302: both displays and the knob, horizontal installation

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Where the digital replication differs from the real unit, this manual says so in the chapter concerned; a full list can be found in Appendix A.

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The X-Plane MD302

The Laminar Research / X-Plane MD302 replication system will be referred to here as the MD302. This has been developed by our team to resemble the Mid-Continent MD302, both in appearance and function. However, the capability and operation of this device in the simulator may and will differ in some areas to the real product.

Just like the real unit, the MD302 is a standby instrument with two small screens and a single push/turn knob. One screen shows attitude, slip and heading, the other one shows airspeed and altitude in a traditional tape format. Everything the pilot does is done with the knob: turn it to set the baro pressure, push it to change the brightness, and hold it down (when the unit is up and running) to open the Options Menu.

Its behavior follows Mid-Continent's own Installation Manual and Operating Instructions (manual number 9017782, revision Q, March 2024): the pilot's sections as well as the installer's Configuration Mode. The battery check and troubleshooting screens follow its sections 5.1.3 and 5.1.4. Where the manual says nothing, the MD302 follows photographs of the real unit.

The unit's memory is a setting file in its folder. Choices made in the Options Menu, as well as in Configuration Mode once they are accepted, are written there at once and are still in place the next time the aircraft is loaded, as on the real unit. Hand-modify the *config.ini* file is NOT recommended unless intensive modification is needed.

Comprehensive comparison can be found towards the end of this manual in Appendix A.

Noticeable additions include:

- **Pop-up window**: a click on either screen of the 3D unit opens a 2D window of the whole unit, knob included. It can be moved, resized and popped out to drag to another display.
- **Heading as shipped**: the heading window shows X-Plane's heading straight away and never loses it, with no MD32 or ARINC source needed. A real unit with no heading source shows "---"; the aircraft's author can switch to the real behaviour in the *config.ini* file.
- **Settings file**: plays the part of both the unit's memory and the configuration module in the real unit's cable harness, this is necessary for the unit inside of a simulated environment to save and load user-specified preferences.
- **Commands**: the knob, the pop-up and Configuration Mode can be bound to keys, joystick buttons or VR controller buttons, in X-Plane dataref format.
- **Instructor failures**: the attitude, airspeed or altimeter can be failed, and any warning message called up, from datarefs. X-Plane's own Failures menu also fails the unit when the author connected it to that side's airspeed indicator and altimeter, or to the AHRS, electric or vacuum gyro. With the default connection it does not.
- **Automatic settings**: panel tilt, panel roll and the airspeed range markings can be taken from the aircraft, so the unit reads correctly without being configured.

The Pop-Up MD302

The pop-up shows the whole front of the unit, both displays and the knob, in a window you can move and size.

Invoking the pop-up MD302

- Click anywhere inside either display of the MD302 in the 3D cockpit, or use the command *MD302 toggle popup window* (assign it to a key or button). The pop-up opens over the cockpit view.
- The knob in the pop-up works like the 3D one. Click its left or right half to turn it, or roll the mouse wheel over it. Click its middle to push it, and hold it there for 2 s to open the Options Menu.





Moving the pop-up MD302

Drag anywhere **except** the knob.

Resizing the pop-up MD302

Drag an edge or a corner of the pop-up. Once it is popped out (see below), resize it like any other window.

Popping out the pop-up MD302

When the mouse pointer is at the top of the pop-up, two click-spots appear in its upper corners.

The icon at the top right puts it in a separate window (for a second display).



The two click-spots: close (red, top left) and pop out (top right)

The popped-out window has its own title bar, *MD302 Popup*. Drag it to another display, or resize and maximise it like any other window.



The pop-up popped out into a window of its own

Closing the pop-up MD302

Move the mouse pointer to the top of the pop-up and click the red click-spot at the top left (shown above). If the pop-up is popped out, use the window's own close button.

It reopens where it was last, as long as X-Plane is running and the aircraft is not reloaded. A pop-up closed while popped out comes back inside X-Plane, not as a separate window.

Vertical installation

- The unit features an **auto-adaptation mechanism** by reading its own placement information in the aircraft's .acf file. Whenever the unit is placed vertically, it turns its pictures to match. To change the orientation, turn the object in Plane Maker; the unit follows it.

The pop-up turns the same way, with the knob at the side, and is sized to fit the screen.



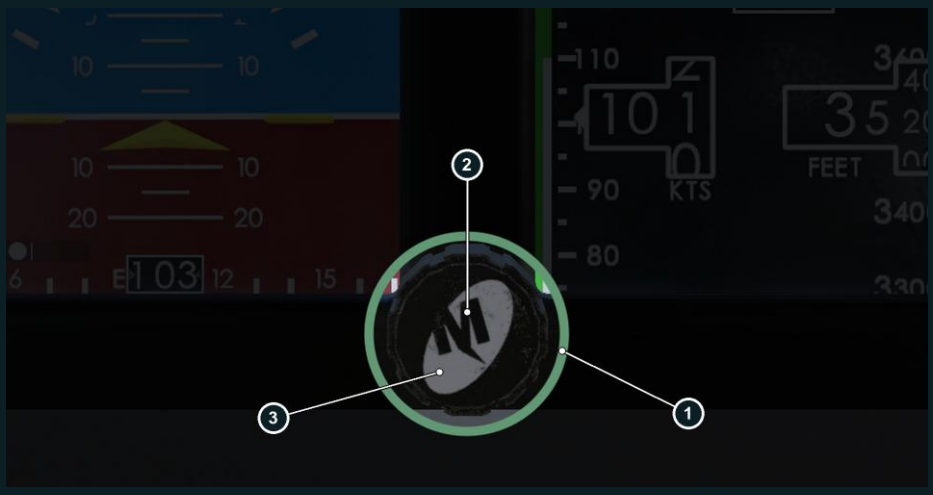
The pop-up of a vertical installation, knob on the right

MD302 Controls and Features

See also: MD302 Installation Manual and Operating Instructions, rev Q, sections 2.1, 4.1, 4.3.1, 4.5.6.

This section identifies the controls and the display features of the MD302. Where relevant, they are discussed in more detail later in this manual.

The control knob



#	Control	What it does
1	Control knob: turn	Sets the baro. In a menu, moves the selection. With the brightness bar shown, changes the brightness. To turn it, click at 3 o'clock (clockwise) or 9 o'clock (counter-clockwise), or roll the wheel.
2	Control knob: push	Click the center. Shows the brightness bar. In a menu it selects, and in a message box it answers.
3	Control knob: push and hold	Hold the center 2 s for the Options Menu.

Important: the MD302 has no power switch. It comes on with aircraft power, and can be turned off from the Options Menu only while it runs on its battery.

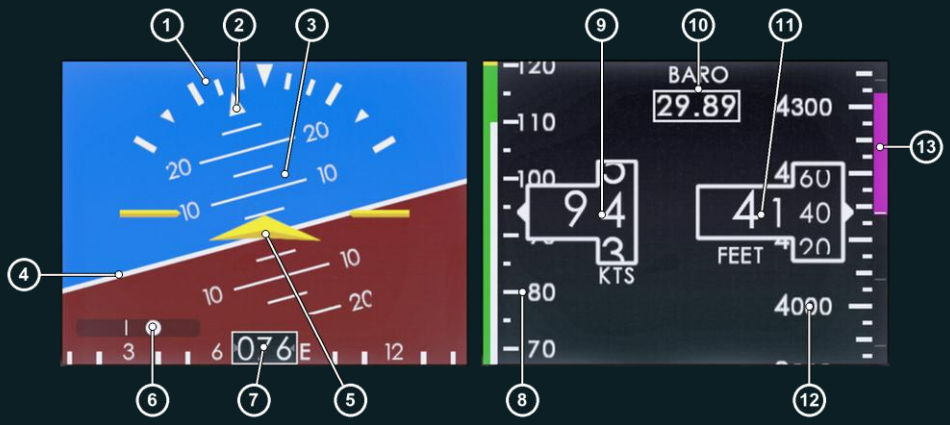
Keys, joystick and VR buttons

- The three commands *MD302 knob push (hold 2 s: Options Menu)*, *MD302 knob turn left* and *MD302 knob turn right* can be assigned to keys, joystick or VR controller buttons in X-Plane's settings.
- Use them in VR and in outside views, where the 3D knob cannot be clicked.

Two more commands help where the knob is hard to use. *MD302 open the Options Menu* opens the menu without holding the knob. *MD302 power-down: remain on (ON)* answers the power-down countdown with ON. All the commands are listed under *Commands* in *Installation and Settings*.

The two displays

The left display shows attitude, slip and heading (items 1–7). The right display shows airspeed, the pressure setting and altitude (items 8–13). In the picture below, the aircraft is climbing in a 15° right turn.



The two displays in flight: attitude on the left, airspeed and altitude on the right

Attitude display (left)

1. Roll scale: the arc at the top, with marks at 10, 20, 30, 45 and 60°.
2. Roll pointer: the small triangle under the scale, turning with the bank. The installer can choose FIXED POINTER instead, where the scale turns.
3. Pitch ladder: a line every 5°, numbered every 10°. As on the real unit, a circle under the roll scale cuts it off.
4. Horizon line: a white line between the blue sky and the dark red ground. The drift and errors of the real unit's own sensors are not simulated: with the default attitude source, the attitude comes from X-Plane.
5. Aircraft symbol: a yellow delta in the center with a yellow bar on each side (DELTA, shown here). SPLIT DELTA or TRADITIONAL can be chosen in the Options Menu.
6. Slip indicator: the ball in the box at the lower left, above the heading tape. Step on the rudder on the side the ball has moved to. With the heading display turned off, it moves to the bottom center.
7. Heading window and tape, along the bottom: the window shows the magnetic heading (076° here). The tape is numbered every 30°, with a letter at the cardinal points (E for 090°). As shipped, the heading is never lost. Loss of the heading source is only simulated when the aircraft's author sets the heading source to ARINC.

At extreme pitch, red chevrons on the ladder point back to the horizon. With the nose 62° up, as in the picture below, they point down.



Red chevrons with the nose 62° up

Airspeed and altitude display (right)

8. Airspeed tape, on the left, with the colored speed bands along its edge: white (flap range), green (normal), yellow (caution) and red from Vne up.
9. Airspeed window: in knots (KTS), or in MPH or KPH if the installer sets AIRSPEED UNITS. The last digit rolls. It reads 0 below the airspeed minimum (30 kt by default, or 20 or 40 kt if the installer sets it).
10. BARO: the pressure setting, at the top (29.89 inHg here).
11. Altitude window: in feet (FEET), or meters (METERS) with ALT UNITS. The last two digits roll in steps of 20 (4,140 ft here).
12. Altitude tape, on the right, numbered every 100 ft (every 100 m with METERS).
13. Altitude trend: the magenta bar beside the altitude tape. It reaches to where the altitude will be in 6 s (ALT TREND in the Options Menu).

Details faithful to the real unit

- Pitch ladder: masked pixel by pixel, as on the unit. It goes on past 90°, with four red chevrons on each side. Its lines are anti-aliased, as in the real unit's screen dumps.



Nose 85° up: the ladder goes on past 90°, back down to 80

- Slip ball: centered between the two lines means coordinated flight (rev Q 4.3.1.3).
- Speed bands come from the aircraft's own V-speeds, or from the installer's RANGE MARKINGS table (full, half or radial marks, in white, green, yellow, red, blue or barber pole).
- Heading: the arrows next to the heading show its source (pointing in for ARINC, pointing out for MD32). With the ARINC source, if it is lost (aircraft power gone or the AHRS failed), the unit runs on its own gyro for 10 s without arrows, then shows "---" (rev Q 4.3.1.2).
- The screen: backlight band, glowing edges, a fine grain and the glass with its own dirt, as on a real TFT behind glass. The real screen's 320 x 240 pixel grid and its glow seen from an angle are only approximated. Its behavior in the cold is not simulated.
- Vertical installation: laid out as in rev Q figure 2.1, with airspeed and altitude side by side, BARO at the top and the start-up page stacked. HORIZONTAL (SWAP) puts the attitude on the right.



A vertical installation in the 3D cockpit: attitude above, airspeed and altitude below

Start-Up

See also: rev Q section 4.2.

At power-up the unit checks its battery (if it has one). It checks its settings (`config.ini`) once, when the aircraft loads. A problem brings up a message box once the flight displays show (see *Failures and Messages*). If the battery has more than 60 minutes left, the battery check shows no message (rev Q 5.1.3). The unit's other power-up self-tests are not simulated.

- With aircraft power on, both displays show the start-up screen: the Mid-Continent logo, SAM | MD302, the software version (S/W VERSION: X-PLANE 2.0.0) and the total run time in hours.
- After about 5 s it switches to the flight displays.
- Aircraft authors: on the ground and on aircraft power, holding the knob down on this screen for 3.5 s opens Configuration Mode (see *Configuration Mode*).



The start-up screen on both displays

Setting the Barometric Pressure

See also: rev Q sections 4.3.1.4, 4.5.4.

An altimeter turns air pressure into altitude, so the pressure setting must be set to the local value before the flight and kept up to date during it.

- Click the knob at 3 o'clock to increase, or at 9 o'clock to decrease. The mouse wheel works too. Each step is 0.01 inHg (or 1 hPa).
- The BARO box turns green while the setting changes, and the altitude follows at once.



BARO turned from 29.89 to 29.90: the box is green

If the installer has turned on CONFIGURE BARO SYNC, the MD302 takes the aircraft's altimeter setting whenever that setting changes. While the two agree, an arrow shows on each side of BARO. A change made on the MD302 holds until the aircraft's setting changes again (rev Q 4.3.1.4).

Important: the MD302 uses the air data source the aircraft's author chose. In some aircraft it shares the main altimeter's setting, so changing one changes the other.

Setting the Units

See also: rev Q sections 4.3.2.2, 4.3.2.3.

The pressure setting can be shown in inches of mercury (IN HG) or millibars (MBAR, the same as hPa), and the altitude in feet or meters.

- Hold the knob 2 s to open the Options Menu.
- Turn to BARO UNITS (or ALT UNITS), push.
- Turn to IN HG or MBAR (FEET or METERS), push.
- Select EXIT MENU, or wait 10 s.

With METERS, the altitude window and tape show meters, so 3,500 ft reads about 1,067 m. FEET under the window becomes METERS.



ALT UNITS open: FEET or METERS; ALT UNITS FEET (default): 3,500 ft; ALT UNITS METERS: the same altitude.

With MBAR, the pressure setting shows in millibars, so 29.89 inHg reads 1012.

Each knob step is then 1 mbar instead of 0.01 inHg.



BARO UNITS open: IN HG or MBAR; BARO UNITS IN HG (default): 29.89; BARO UNITS MBAR: the same setting, 1012

Adjusting Brightness

See also: rev Q sections 4.3.3, 4.5.5.

The displays dim and brighten on their own with the light in the cockpit. Set them by hand when they are too bright, for example at night. In daylight they are already at full brightness, so they can only be dimmed.

- To change the brightness:
  1. Push the knob. A green brightness bar appears along the left edge of the attitude display.
  2. Turn clockwise for brighter, counter-clockwise for dimmer.
  3. The bar goes away 4 s after the last change.
- The bar works between the lowest point of the dimming curve and full brightness. At night the photocell has already turned the backlight part of the way down (about 42 % at 21:00, see below). Fewer steps are then left to dim, but the displays can still be dimmed further or turned up. They are already at the lowest point only in full darkness.



The brightness bar

Each knob step changes the brightness by about 9 %. In daylight, about 28 steps down from the top reach the lowest point of the dimming curve (10 % with the default curve). The displays never go fully dark.



Just shown: full brightness

12 steps counter-clockwise



24 steps counter-clockwise

36 steps: the lowest point (reached at 28)

Brightness bar just shown: full brightness; 12 steps counter-clockwise; 24 steps counter-clockwise; 36 steps counter-clockwise: no dimmer than at 28 steps, the lowest point

How automatic dimming works

- A photocell follows X-Plane's light in the cockpit. The installer's dimming curve turns that light into a backlight level.
- In aircraft where the author linked it, the panel-light knob dims the MD302 instead (external dimming, 5 / 14 / 28 V). When the knob is turned right down, the photocell takes over again.
- Your own setting cannot go below the lowest point the installer set. It is cleared when the unit turns off, and at the next start the displays follow the dimming curve again.

The pictures below show the same parked aircraft at four times of day, with INTERNAL (the default). As the light fades, the photocell moves the backlight level down the dimming curve. Each caption gives that level. X-Plane 12 adjusts its camera exposure to the light, and the default curve is tuned for that. So the displays look at least as bright at night as by day, while the panel around them goes dark. A level of 42 % at night is therefore not dim, unlike the panel-light knob at 40 % with EXT 28V below.



15:30, daylight: 100 %



17:30, sunset: about 85 %



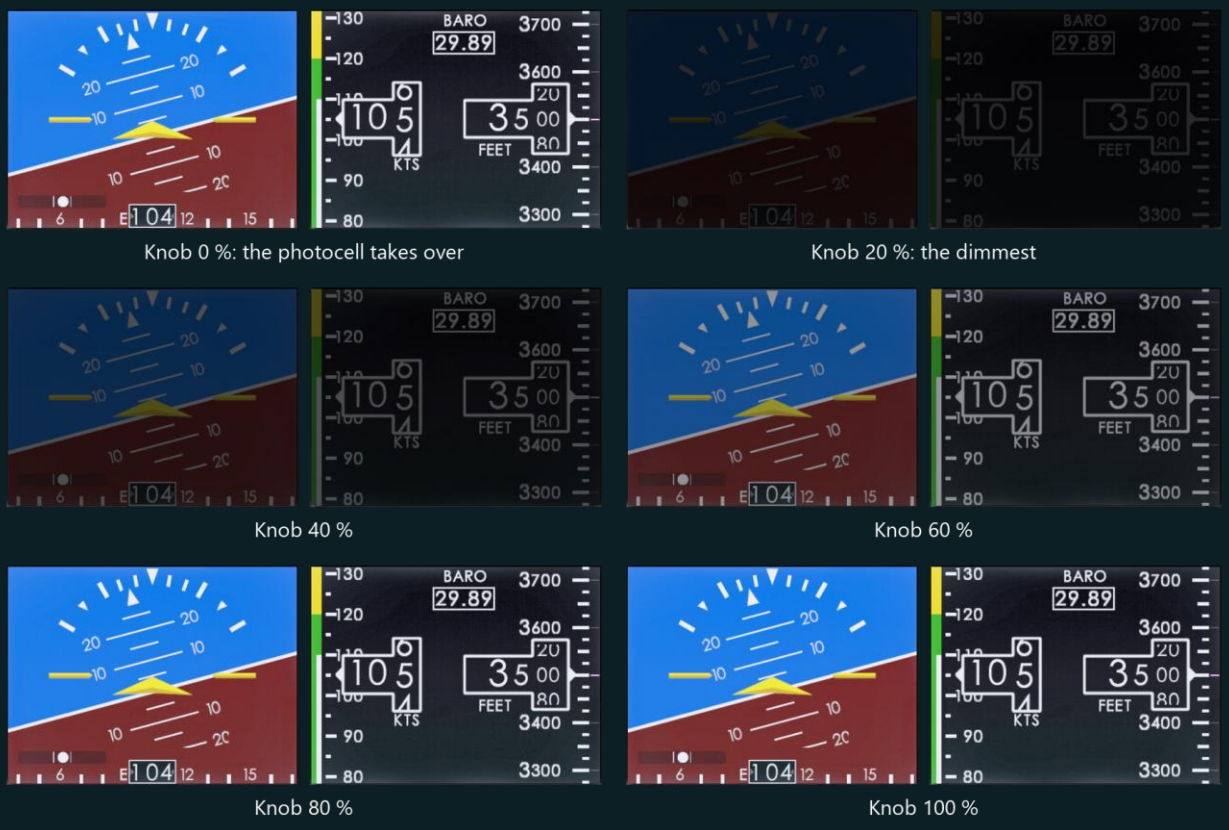
18:30, dusk: about 48 %



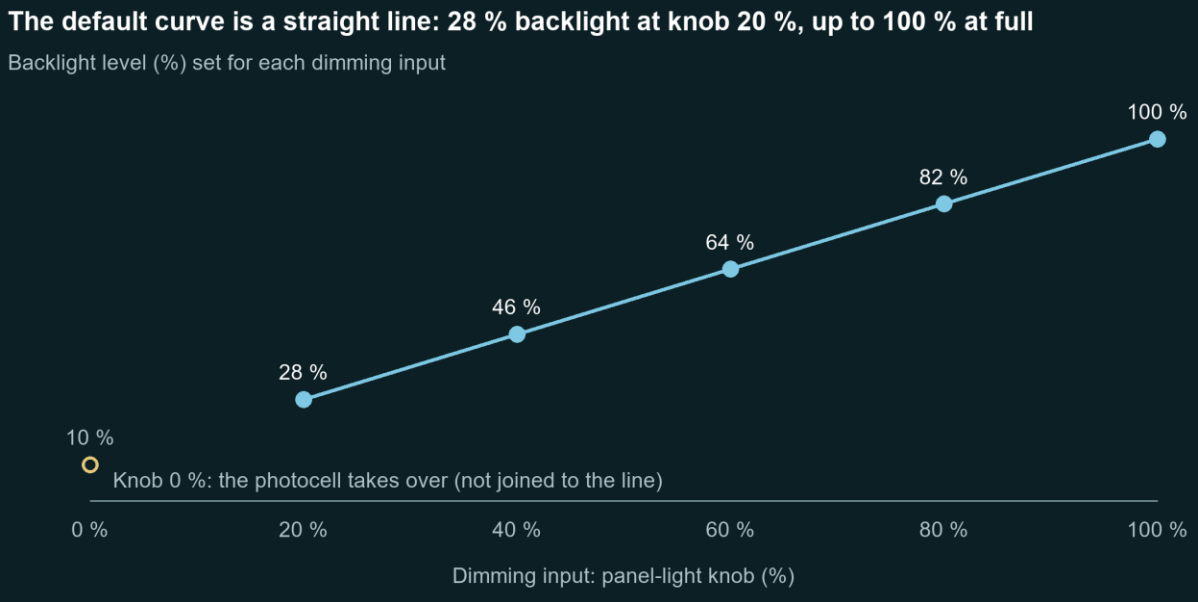
21:00, night: about 42 %

INTERNAL, 15:30, daylight: backlight level 100 %; INTERNAL, 17:30, sunset: backlight level about 85 %. The low sun lights up the marks on the right display's glass; INTERNAL, 18:30, dusk: backlight level about 48 %; INTERNAL, 21:00, night: backlight level about 42 %. The panel is dark, but the displays are still easy to read

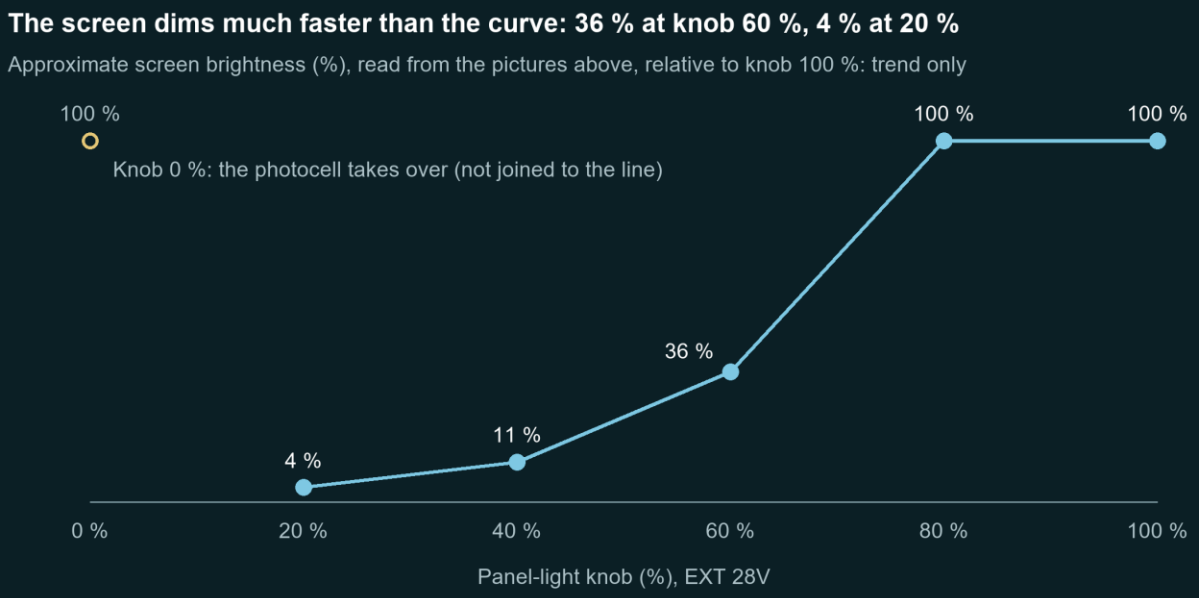
With external dimming (here EXT 28V), the panel-light knob sets the backlight. The displays are dimmest just above 0 %, still dim at 20 %, and brighten as the knob goes up. At 80 % and 100 % they are as bright as the photocell makes them in daylight. At 0 % they are bright again. At or below the installer's cut-off on the dimming input (0 by default, set under CONFIGURE DIMMING), the photocell takes over, and in daylight that is bright.



EXT 28V, panel-light knob at 0 %: the photocell takes over; EXT 28V, panel-light knob at 20 %: the dimmest of these six; EXT 28V, panel-light knob at 40 %; EXT 28V, panel-light knob at 60 %; EXT 28V, panel-light knob at 80 %; EXT 28V, panel-light knob at 100 %



The default curve. The installer sets it in Configuration Mode > CONFIGURE DIMMING



Approximate, read from the six EXT 28V pictures above in daylight. It shows the trend only

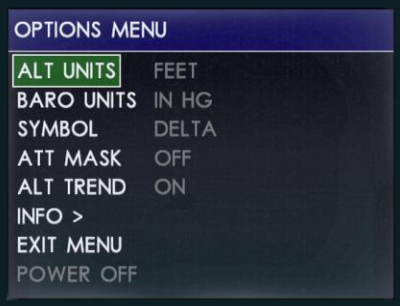
The curve is a straight line, but the screen's response is not. Below 60 %, each 20 % of knob travel leaves about half the brightness, so the low end is much darker than the line suggests. In daylight the screen reaches its full brightness at a backlight level of about 78 %, so knob 80 % and 100 % look the same.

The Options Menu

See also: rev Q section 4.3.2.

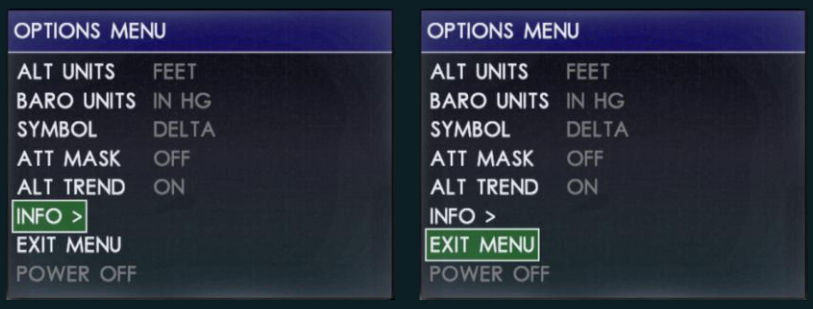
- Hold the knob for 2 s. The menu replaces the attitude display.
- Turn to move, push to select. For a list of values, turn to the value and push again.
- EXIT MENU closes the menu. It also closes after 10 s without activity.
- What you pick is kept for the next flight in this aircraft.
- Menu colors: white items can be selected and grey ones are for information only. An item with “>” opens a sub-menu (rev Q 4.3.2.1).
- A value is saved the moment you pick it, even if you never select EXIT MENU.

The cursor is a green box. Each item’s current value shows in grey beside it. POWER OFF is grey while the unit is on aircraft power.



The Options Menu, cursor on ALT UNITS

Item	What it does
ALT UNITS	FEET or METERS
BARO UNITS	IN HG or MBAR
SYMBOL	DELTA, SPLIT DELTA or TRADITIONAL
ATT MASK	ON darkens the corners so it looks round
ALT TREND	Altitude trend bar on / off
INFO >	Opens the INFO Menu
EXIT MENU	Closes the menu
POWER OFF	Turns the unit off, only while on the battery (grey on aircraft power)



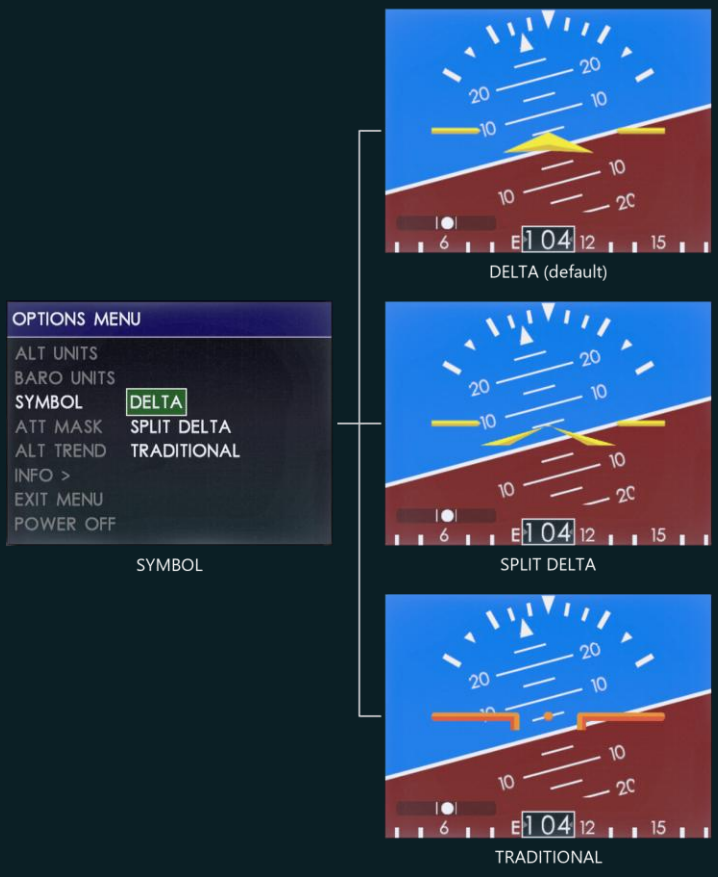
Cursor on INFO > Cursor on EXIT MENU

Cursor on INFO >: push to open the INFO Menu; Cursor on EXIT MENU: push to close the menu

The aircraft symbol (SYMBOL)

The symbol in the center of the attitude display stands for your aircraft.

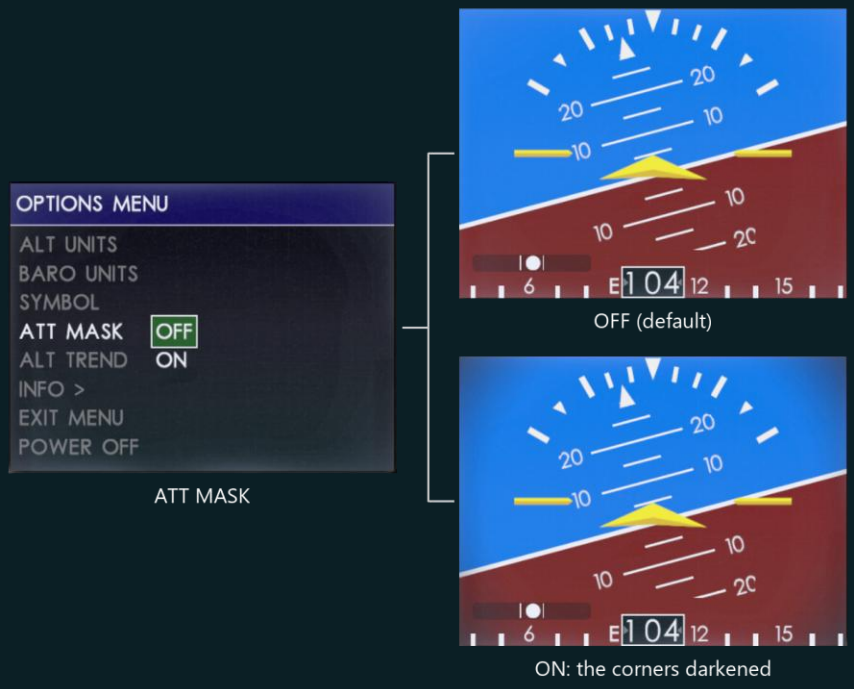
DELTA is a yellow delta. SPLIT DELTA splits it into a left and a right half. TRADITIONAL is the older orange symbol: a bar on each side and a dot in the center.



SYMBOL open: DELTA, SPLIT DELTA or TRADITIONAL; SYMBOL DELTA (default); SYMBOL SPLIT DELTA; SYMBOL TRADITIONAL

The attitude mask (ATT MASK)

ON darkens the four corners of the attitude display, sky and ground alike, so the eye goes to the center. The center is unchanged.



ATT MASK open: OFF or ON; ATT MASK OFF (default); ATT MASK ON: the corners darkened

The altitude trend (ALT TREND)

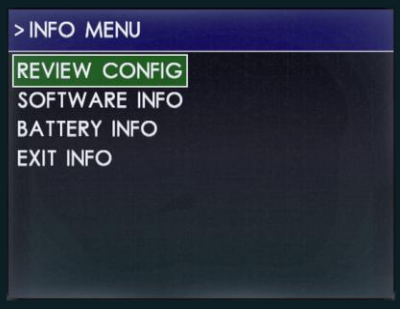
ON shows a magenta bar beside the altitude tape, reaching to where the altitude will be in 6 s. When climbing at about 750 fpm, the bar reaches about 75 ft above the current altitude. OFF removes the bar.



ALT TREND open: OFF or ON; ALT TREND ON (default), climbing; ALT TREND OFF: no trend bar

The INFO Menu

Item	What it shows
REVIEW CONFIG	Every setting the aircraft's installer made
SOFTWARE INFO	The software version: X-PLANE 2.0.0 (with an MD32 heading source, also MD32 SW VERSION)
BATTERY INFO	EST RUN TIME (MINS), TEMPERATURE (C), CHARGE (%), CAPACITY (MAH) and, scrolled down, TYPE. Grey on a unit without a battery
EXIT INFO	Back to the Options Menu

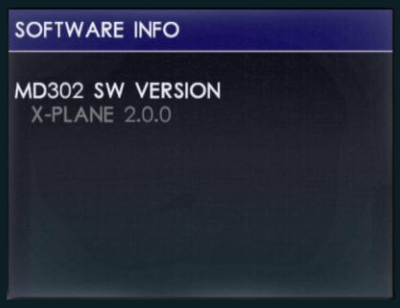


The INFO Menu

REVIEW CONFIG lists the installer's settings, 8 lines to a screen. Each knob step scrolls one line, and the 53 lines take 7 screens (one more line if an SSEC file is loaded). The INFO pages only show values, and nothing can be changed there.

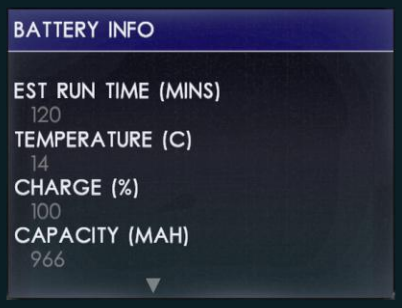


REVIEW CONFIG snapshots.

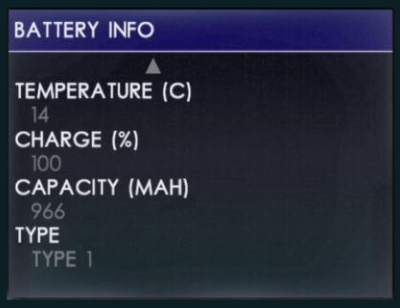


SOFTWARE INFO: MD302 SW VERSION X-PLANE 2.0.0 (xLua 2.0)

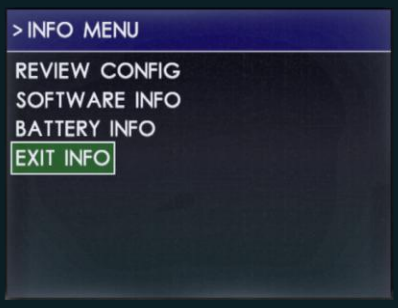
On a full battery, BATTERY INFO shows about 120 minutes of run time. The temperature follows the outside air, held between 5 and 40 °C. Battery heating and ageing are not simulated.



BATTERY INFO, full battery



Scrolled down: the TYPE



Cursor on EXIT INFO

BATTERY INFO on a full battery; BATTERY INFO scrolled down: the battery TYPE; Cursor on EXIT INFO: push to go back to the Options Menu

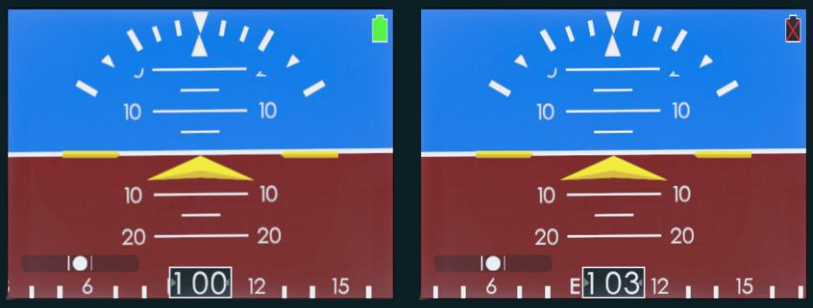
Battery Operation

See also: rev Q sections 4.4.1, 4.4.2, 5.1.3.

- The switch to the battery is automatic, with no pilot action needed (rev Q 4.4.1).
- Below 20 % charge the icon turns black with a red X. With the default 120-minute battery, that leaves about 24 minutes. On the real unit the red X means that as little as 10 minutes may be left (rev Q 4.4.1).
- If the installer turned AUTO-OFF off, turn the MD302 off yourself after the flight (POWER OFF), or the battery runs down (rev Q 4.4.2).
- The aircraft's author chooses the battery: lithium-ion (default), nickel metal hydride, or none (the unit then goes off with aircraft power).

The charge is kept from one flight to the next. A battery run down in one flight is still low when the aircraft is loaded again, and the start-up check then warns if less than 60 minutes are left.

Aircraft power lost in flight: the unit keeps running on its battery, and a green battery icon shows at the top right of the attitude display.



On the battery: green icon — Below 20 %: black icon, red X

On the battery in flight: the green battery icon; Below 20 % charge: the black battery icon with a red X

On the ground

- When the unit is on the battery with no airspeed (indicated airspeed below 20 kt, so taxiing counts too) and AUTO-OFF is on, a *POWER DOWN IN 60 SEC* box appears over the attitude display and counts down, with OFF lit first. If airspeed returns, the box goes away; when the airspeed is gone again, the count starts again from 60.
  - ON keeps it running on the battery.
  - OFF, or the count reaching 0, turns it off.
- POWER OFF in the Options Menu turns it off any time it is on the battery, in the air too.
- When aircraft power comes back, the unit switches back to it and recharges its battery.



The power-down countdown, OFF lit

Units without a battery

Some MD302 versions have no battery. The aircraft's author chooses one with *battery_type = none* in *config.ini*. Such a unit runs on aircraft power only:

- It turns off as soon as aircraft power goes, on the ground or in the air. There is no POWER DOWN IN 60 SEC box and no battery icon.
- It starts again when aircraft power comes back.
- At start-up there is no battery check and no battery warning.
- BATTERY INFO in the INFO Menu and BATTERY STORAGE MODE in Configuration Mode are grey. POWER OFF in the Options Menu stays grey too, since it works only on the battery.
- An instructor's *fail_message* 12 (BATTERY OVER TEMPERATURE) is ignored.

Failures and Messages

See also: rev Q sections 1.2, 5.1.4.

Failures

- A failed sensor puts a red X over the part it feeds (attitude, airspeed or altitude); the rest keeps working. A failed attitude display shows only the red X on black, with no battery icon, brightness bar or power-down box. A power-down countdown still runs, out of sight, and switches the unit off at 0.
- Leaving the sensor range also fails a sensor. Attitude fails when pitch or roll is faster than 300°/s, and comes back 3 s after the rate is back in range. Altitude fails below -1,500 ft or above 55,000 ft pressure altitude (at the standard setting, 29.92 inHg). Airspeed fails above 500 kt.



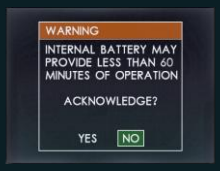
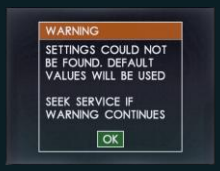
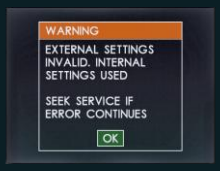
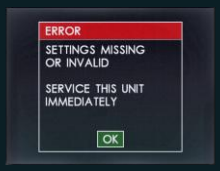
Attitude failed: a red X over the attitude display; Airspeed and altitude failed: a red X over each tape, BARO still shown

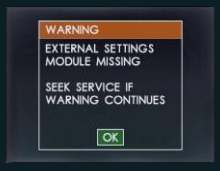
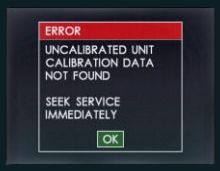
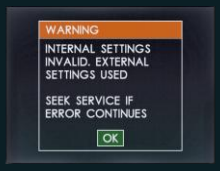
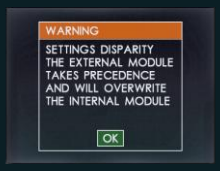
X-Plane's Failures Menu

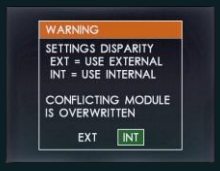
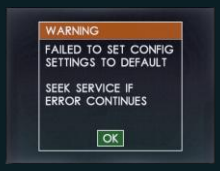
- Failing an instrument the MD302 reads (Flight > Failures) gives the same red X. Which instruments count depends on how the aircraft's author connected the MD302: that side's airspeed indicator and altimeter, and the AHRS, the electric gyro or the vacuum gyro (Artificial horizon in X-Plane's Failures menu).
- With the default connection there is no matching item in the menu. Use the datarefs below instead.
- A blocked pitot or static port is not a failure: the unit shows the wrong values, as a real one would.

Messages

- Some situations put a message box over the attitude display, for example when the battery may last less than 60 minutes.
- Push the knob to answer. For YES / NO, turn to the answer first, then push.
-

Message	What it means	What to do
INTERNAL BATTERY MAY PROVIDE LESS THAN 60 MINUTES OF OPERATION  *The battery warning at start-up, NO lit*	Less than 60 minutes of battery left at power-up	ACKNOWLEDGE? Turn to YES or NO and push. Either answer closes the box. Backup time on the battery may be short
SETTINGS COULD NOT BE FOUND  *SETTINGS COULD NOT BE FOUND, OK lit*	`config.ini` is missing	OK; the next save writes a new one
EXTERNAL SETTINGS INVALID  *EXTERNAL SETTINGS INVALID, OK lit*	`config.ini` was damaged and put back from its backup	OK
SETTINGS MISSING OR INVALID  *SETTINGS MISSING OR INVALID, OK lit*	A value was not allowed and was replaced. A number with limits goes to the nearest limit, PANEL TILT or PANEL ROLL to auto, AIRSPEED MINIMUM to 30, a bad dimming curve to the default curve, a bad VNE table to none, anything else to its default	OK; see Log.txt
BATTERY OVER TEMPERATURE  *BATTERY OVER TEMPERATURE, OK lit*	The battery is too hot. On the battery the unit turns off at once. On aircraft power it keeps running, but turns off if aircraft power goes, and the battery icon stays black with a red X until the aircraft is reloaded. Instructor only (`fail_message` 12)	OK (only on aircraft power)

Message	What it means	What to do
EXTERNAL SETTINGS MODULE MISSING  *EXTERNAL SETTINGS MODULE MISSING, OK lit*	The configuration module is missing or has failed, so the unit uses its internal settings. Instructor only (*fail_message* 2)	OK; seek service if it continues
UNCALIBRATED UNIT  *UNCALIBRATED UNIT, OK lit*	Calibration data not found. Instructor only (*fail_message* 4)	OK; the box says to seek service immediately
DETECTED STARTUP FAILURE  *DETECTED STARTUP FAILURE, OK lit*	An internal failure at start-up. Instructor only (*fail_message* 5)	OK; the box says to service the unit immediately
INTERNAL SETTINGS INVALID  *INTERNAL SETTINGS INVALID, OK lit*	The unit's own settings were invalid, so the configuration module's settings are used. Instructor only (*fail_message* 6)	OK; seek service if it continues
SETTINGS DISPARITY (OK)  *SETTINGS DISPARITY, OK lit*	The unit's and the module's settings differ, typically after a unit is replaced. The module's settings win. Instructor only (*fail_message* 8)	OK

Message	What it means	What to do
SETTINGS DISPARITY (EXT / INT)  *SETTINGS DISPARITY with EXT and INT, INT lit*	As above, but in Configuration Mode, where you choose which settings to keep. Instructor only (*fail_message* 9)	Turn to EXT or INT and push. In the simulator either answer only closes the box
FAILED TO SET CONFIG SETTINGS TO DEFAULT  *FAILED TO SET CONFIG SETTINGS TO DEFAULT, OK lit*	In Configuration Mode, the settings were lost and could not be reset. Instructor only (*fail_message* 10)	OK. Restart and configure again, and seek service if it continues
Red X (attitude, airspeed, altitude)	That sensor has failed	Out of the sensor range, it clears by itself once back in range (attitude 3 s later). Set from a dataref or X-Plane's Failures menu, it stays until cleared. The rest keeps working

Failures for instructors

Dataref	Effect
laminar/md302/fail_attitude	1 = red X on attitude; 0 = working
laminar/md302/fail_airspeed	1 = red X on airspeed; 0 = working
laminar/md302/fail_altimeter	1 = red X on altitude; 0 = working
laminar/md302/fail_message	1–12 shows that message box, then reads 0. The box waits for flight mode; in Configuration Mode only 9–11 show. On the battery, 12 turns the unit off with no box

Message numbers: 1 battery test, 2 settings module missing, 3 settings missing or invalid, 4 uncalibrated unit, 5 start-up failure, 6 internal settings invalid, 7 external settings invalid, 8 and 9 settings disparity, 10 failed to set defaults, 11 settings not found, 12 battery over temperature.

Configuration Mode (for Aircraft Authors)

See also: rev Q sections 4.5, 5.1.2.

Configuration Mode is where the installer sets the unit up for the aircraft: speed markings, dimming, orientation, the heading display and more. The X-Plane MD302 has it too. The heading source is set only in `config.ini` (`heading`).

Entering Configuration Mode

- On the ground, on aircraft power: **push and hold** the knob while the start-up screen shows.
- After half a second the attitude display shows CONTINUE HOLDING TO ENTER CONFIG MODE. After 3.5 s the CONFIGURE MENU opens by itself. If you let go earlier, the unit starts as usual.
- Or, with the unit on and on aircraft power, use the command *MD302 restart in Configuration Mode (on the ground)*.

  The real unit's way also works. Hold the knob while aircraft power comes on. CONTINUE HOLDING appears after about 6 s, and the introduction screen after about 9 s. Then let go to open the CONFIGURE MENU. In the simulator one mouse cannot do both at once, so the start-up screen route above was added.



Knob held on the start-up screen: CONTINUE HOLDING; The CONFIG MENU on the right, a live preview on the left

If aircraft power goes or the aircraft moves while the knob is held, the unit starts as usual. On the battery or in motion, a push held through the start-up screen does nothing. It does not open the Options Menu either. Once in Configuration Mode, a unit with charge left in its battery stays there if aircraft power goes. It runs on the battery, with no power-down countdown. Rev Q only implies this: its section 4.5.10 makes UPDATE SOFTWARE unavailable on battery power.

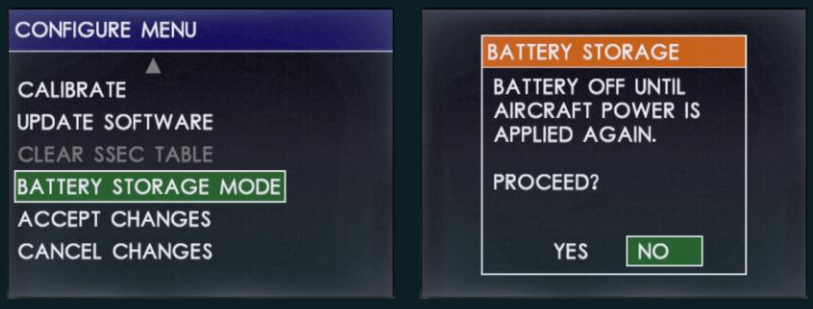
The CONFIGURE MENU

The menu is on the airspeed and altitude display, and the attitude display shows a live preview of the setting being changed. Turn to move the green cursor, and push to open an entry. An arrow at the top or bottom of the list means the list goes on in that direction.

Entry	What it sets
CONFIGURE AIRCRAFT	Panel tilt / roll, Mmo, range markings, VNE table
CONFIGURE ARINC	Pass-through, transmit speed (shown only)
CONFIGURE BARO SYNC	Baro sync
CONFIGURE DIMMING	Dimming control and curve
CONFIGURE DISPLAY	Roll display, airspeed units, airspeed minimum, orientation, horizon
CONFIGURE HEADING	Heading display on / off
CONFIGURE AUTO-OFF	Power-down countdown on the ground on / off
CALIBRATE	Shown only; CALIBRATE HEADING grey
UPDATE SOFTWARE	Always ends in FAILED UPDATE (no USB)
CLEAR SSEC TABLE	Empties the SSEC table name once you answer YES at PROCEED? and then select ACCEPT CHANGES (grey when none is named)
BATTERY STORAGE MODE	Battery off until aircraft power returns
ACCEPT / CANCEL CHANGES	Save to *config.ini* and restart / restart without saving

The list has 13 entries, six to a screen. CLEAR SSEC TABLE can be selected only when *config.ini* names an SSEC table. Otherwise it is grey and the cursor skips it. The table name shows on the start-up screen and in REVIEW CONFIG, and no correction is applied.

BATTERY STORAGE MODE asks for confirmation first. YES turns the battery off until aircraft power is applied again, for when the aircraft is put away. The entry is grey while there are unsaved changes, and on a unit without a battery.

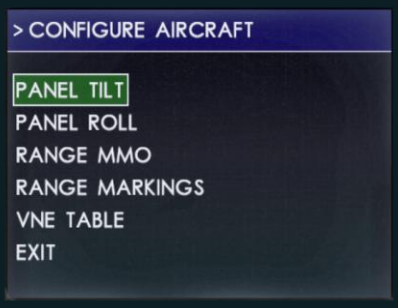


The end of the list The BATTERY STORAGE box

The end of the list: CLEAR SSEC TABLE grey, cursor on BATTERY STORAGE MODE; The BATTERY STORAGE box, cursor on NO. YES turns the battery off until aircraft power returns

Configure Aircraft

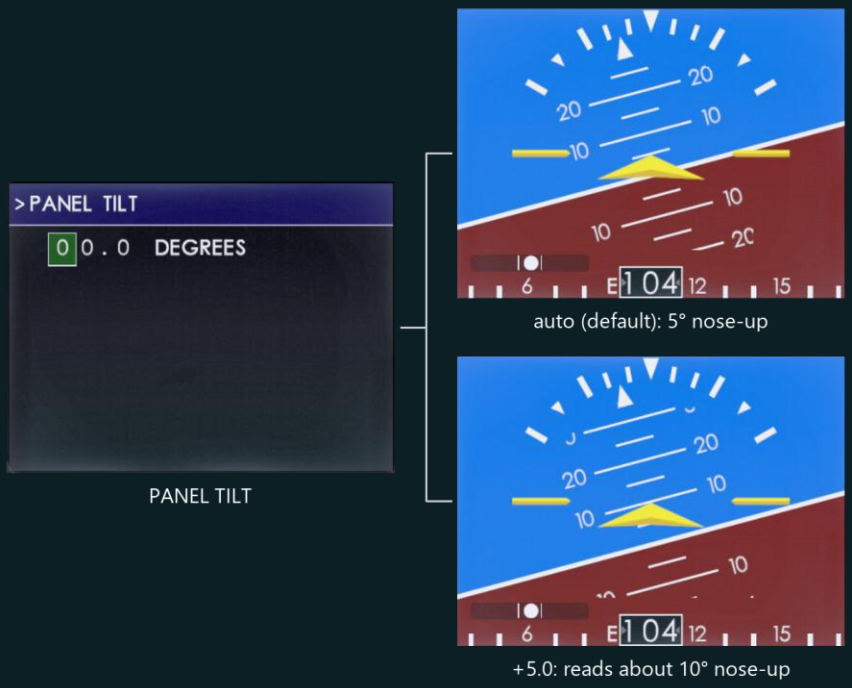
These settings cover how the unit sits in the panel, and the markings and limits on the airspeed tape.



CONFIGURE AIRCRAFT: PANEL TILT, PANEL ROLL, RANGE MMO, RANGE MARKINGS, VNE TABLE

PANEL TILT: how far the panel leans back, -90 to +90° in 0.1° steps. Push to move to the next digit. The last position is the sign. By default it is taken from how the unit sits in the aircraft.

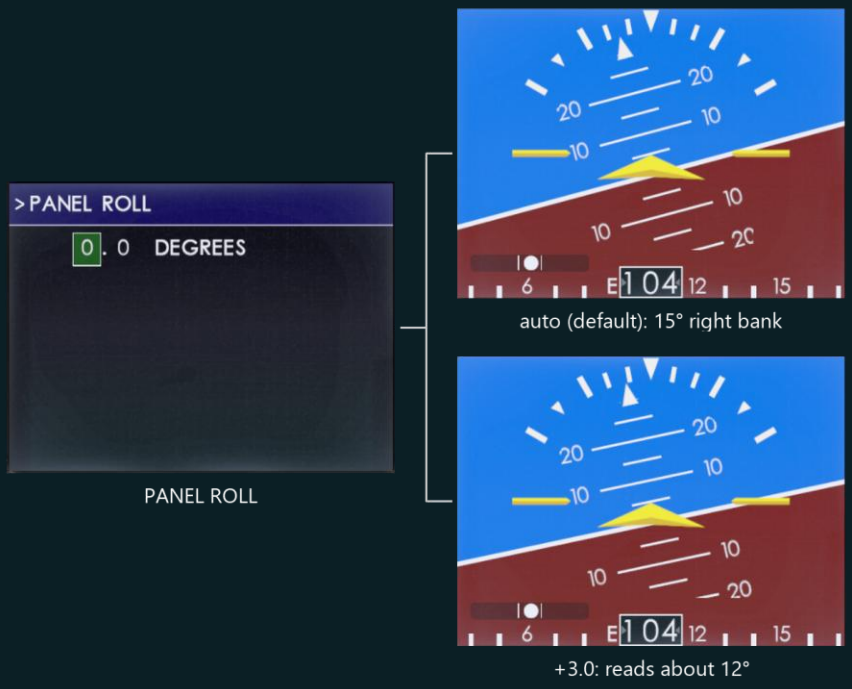
Set right, it matches the installation. Set wrong, the attitude is off. With 5° too much, the same attitude shows 5° more nose-up, so 5° up reads about 10° and the horizon sits one ladder line lower.



PANEL TILT: 0.0 degrees, cursor on the first digit; PANEL TILT auto (default): 5° nose-up; PANEL TILT +5.0: the same attitude reads about 10° nose-up

PANEL ROLL: the same sideways, -5 to +5° in 0.5° steps.

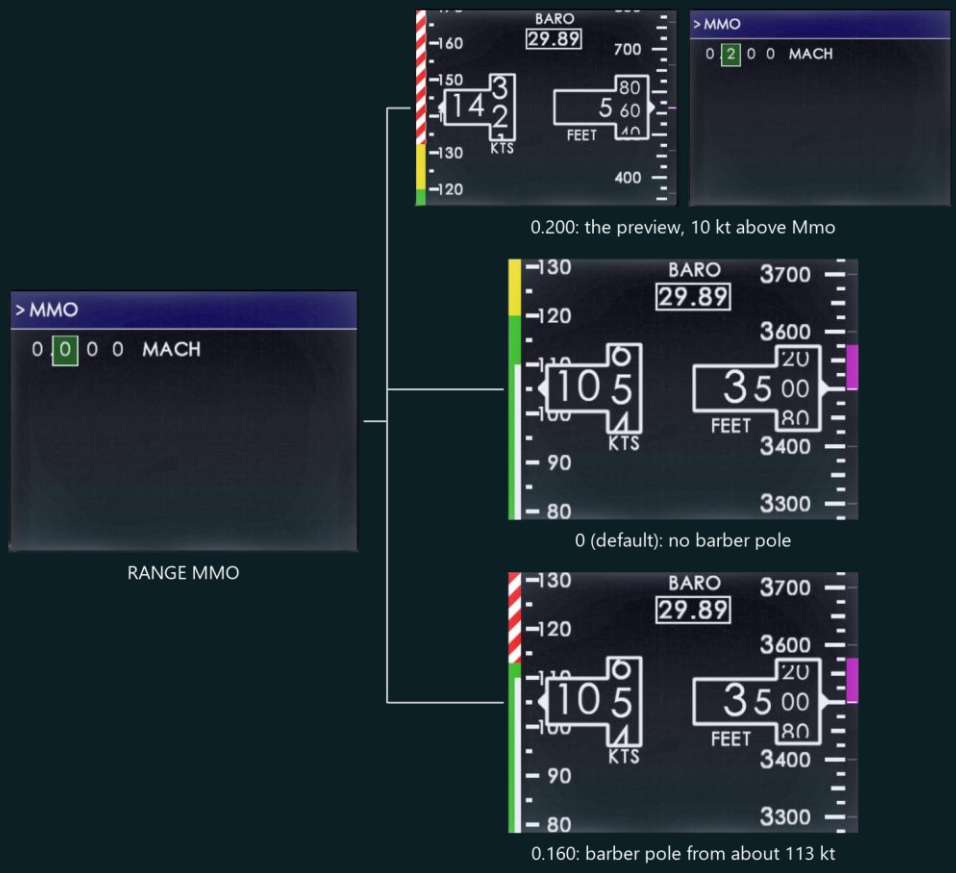
With +3.0, every roll reads 3° to the left: a 15° right bank reads about 12°, and level flight shows a 3° left bank.



PANEL ROLL: 0.0 degrees; PANEL ROLL auto (default): 15° right bank; PANEL ROLL +3.0: the same bank reads about 12°

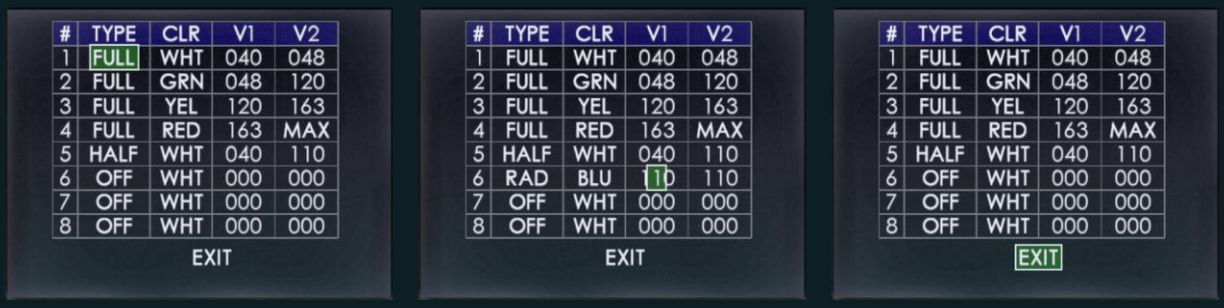
RANGE MMO: the maximum Mach number, 0.000 to 0.999. At 0 no barber pole is drawn. While it is being set, the preview changes to the airspeed display.

With an Mmo set, a red-and-white barber pole marks the airspeed tape from the speed where the Mach number reaches Mmo (airspeed × Mmo ÷ Mach). Below about Mach 0.05 (standing or taxiing) the pole is not drawn. At Mach 0.148 and 105 kt, an Mmo of 0.160 starts it at about 113 kt.



MMO: 0.000 MACH; RANGE MMO at 0.200: the preview puts the airspeed 10 kt above the Mmo speed (132 kt at sea level), so the barber pole shows; RANGE MMO 0 (default): no barber pole (climbing, so the altitude trend bar shows); RANGE MMO 0.160: barber pole from about 113 kt (climbing, so the altitude trend bar shows)

RANGE MARKINGS: the colored bands along the airspeed tape. There are eight lines. Each has a type (FULL, HALF, RAD or OFF), a color, and the speeds where it starts and ends (V1, V2). By default the first five come from the aircraft's V-speeds (here Vso 40, Vs 48, Vfe 110, Vno 120, Vne 163). Each push moves the cursor one step, and the speeds are set one digit at a time. With the default table, 43 pushes reach EXIT. Once a line is changed, the table is saved as it stands and no longer follows the aircraft's V-speeds.



#	TYPE	CLR	V1	V2
1	FULL	WHT	040	048
2	FULL	GRN	048	120
3	FULL	YEL	120	163
4	FULL	RED	163	MAX
5	HALF	WHT	040	110
6	OFF	WHT	000	000
7	OFF	WHT	000	000
8	OFF	WHT	000	000

EXIT

#	TYPE	CLR	V1	V2
1	FULL	WHT	040	048
2	FULL	GRN	048	120
3	FULL	YEL	120	163
4	FULL	RED	163	MAX
5	HALF	WHT	040	110
6	RAD	BLU	110	110
7	OFF	WHT	000	000
8	OFF	WHT	000	000

EXIT

#	TYPE	CLR	V1	V2
1	FULL	WHT	040	048
2	FULL	GRN	048	120
3	FULL	YEL	120	163
4	FULL	RED	163	MAX
5	HALF	WHT	040	110
6	OFF	WHT	000	000
7	OFF	WHT	000	000
8	OFF	WHT	000	000

EXIT

RANGE MARKINGS: the default lines, from the aircraft's V-speeds; RANGE MARKINGS with line 6 set to RAD BLU 110; RANGE MARKINGS: cursor on EXIT

A RAD line is a single radial line. Line 6 set to RAD BLU 110 draws a blue line across the bands at 110 kt.



RANGE MARKINGS default (the aircraft's V-speeds); RANGE MARKINGS with RAD BLU 110 added; Only the speed tape: the blue line at 110 kt

VNE TABLE: a never-exceed speed that changes with altitude. Up to 7 points (altitude, speed) are joined in order of altitude, and the first is at 0 ft. Above that speed, a barber pole shows. The table is empty by default.

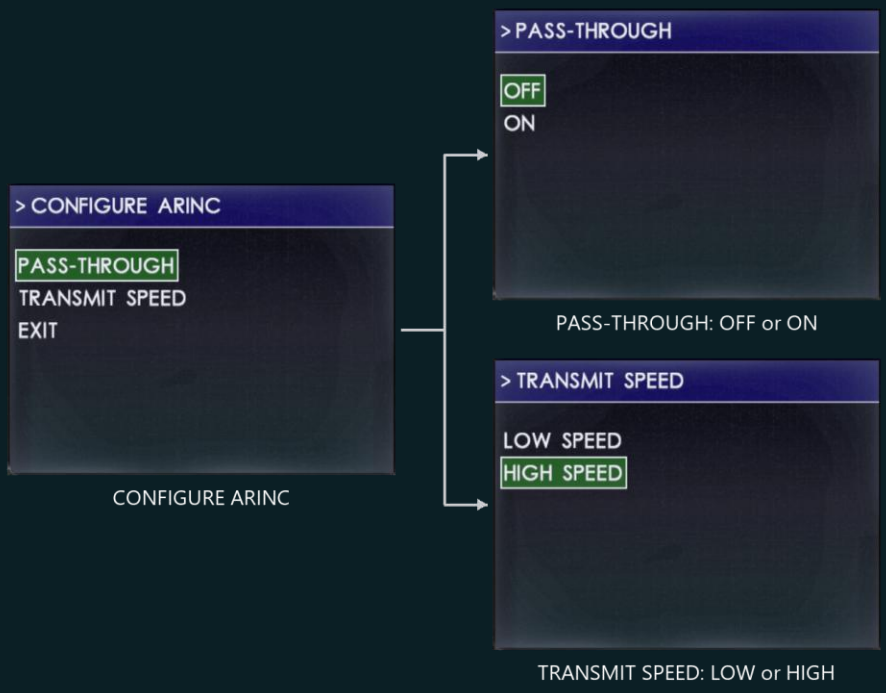
With 115 kt at 0 ft, the barber pole starts at 115 kt, just below the yellow band.



VNE TABLE: empty; VNE TABLE point 1 set to 100 kt: the preview puts the airspeed 10 kt above it, and the barber pole starts at 100 kt; VNE TABLE empty (default); VNE TABLE 0 ft, 115 kt: barber pole from 115 kt

Configure ARINC

X-Plane does not read the MD302's ARINC 429 output. Both settings are only saved and shown in REVIEW CONFIG, and nothing on the displays changes.

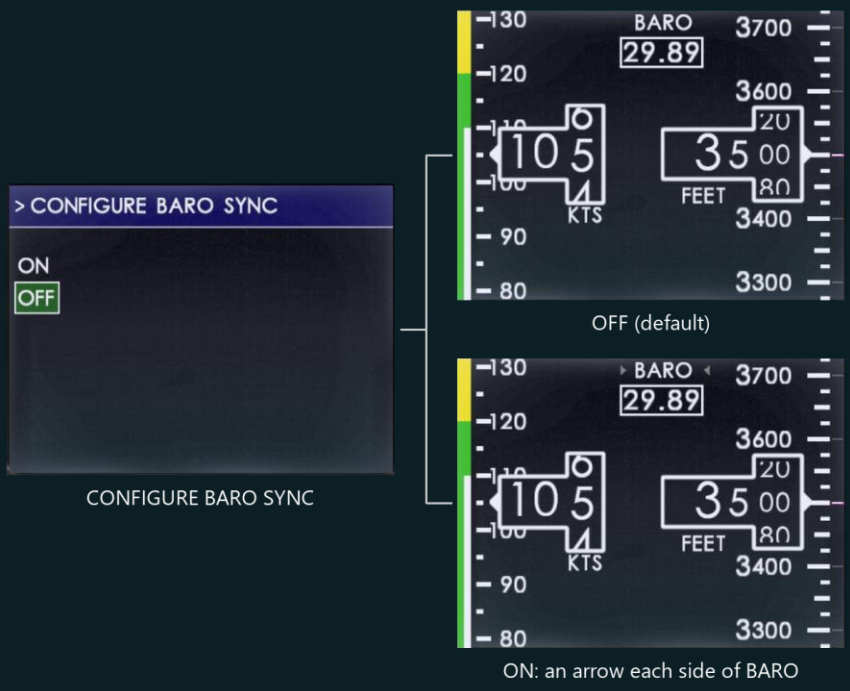


CONFIGURE ARINC: PASS-THROUGH, TRANSMIT SPEED; PASS-THROUGH: OFF or ON; TRANSMIT SPEED: LOW SPEED or HIGH SPEED

Configure Baro Sync

ON makes the MD302 take its pressure setting from the aircraft's main altimeter.

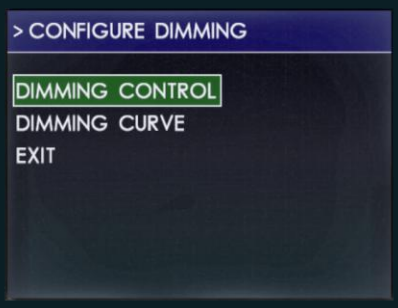
While the two are in sync, a small grey arrow shows on each side of BARO. Turning the MD302's own knob removes the arrows until the main altimeter's setting is changed again.



CONFIGURE BARO SYNC: ON or OFF; BARO SYNC OFF (default); BARO SYNC ON: an arrow on each side of BARO

Configure Dimming

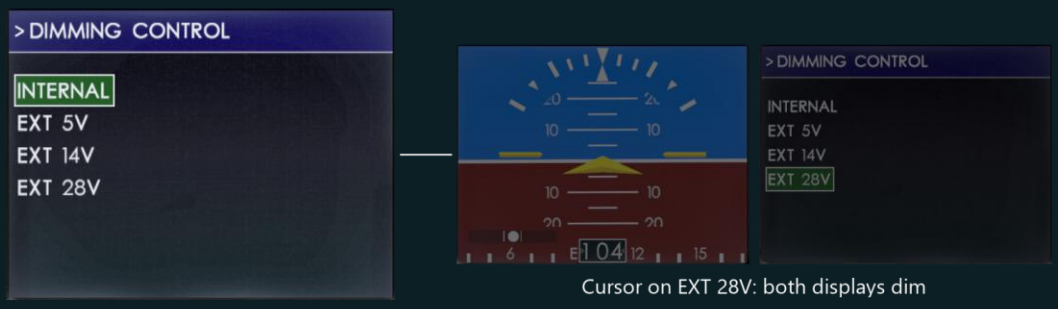
DIMMING CONTROL picks what sets the backlight. DIMMING CURVE sets how bright the displays are at each input.



CONFIGURE DIMMING: DIMMING CONTROL, DIMMING CURVE

INTERNAL follows the unit's photocell, which reads the light in X-Plane's cockpit. EXT 5V, EXT 14V and EXT 28V follow the aircraft's panel-light knob.

With the cursor on EXT 28V, both displays in the preview dim at once to the panel-light knob's setting (here 50 %).



DIMMING CONTROL

Cursor on EXT 28V: both displays dim

DIMMING CONTROL: INTERNAL, EXT 5V, EXT 14V or EXT 28V; Cursor on EXT 28V: both displays dim to the panel-light knob

DIMMING CURVE has six points (PT 1 to PT 6 in REVIEW CONFIG), giving the backlight level at 0, 20, 40, 60, 80 and 100 % input. By default it is a straight line from 10 to 100 %. Turn to change the current point, and push to go to the next. Raising or lowering a point makes the displays brighter or dimmer at that input.

With external dimming a seventh step follows, a vertical line with a dot (PT 7 in REVIEW CONFIG). At or below this input, the photocell takes over. By default it is 0, the input with the knob turned fully down or with no power on the lighting bus.



INTERNAL: MIN, AMBIENT, MAX light

EXT 28V: 0 to 28 V, line = knob now

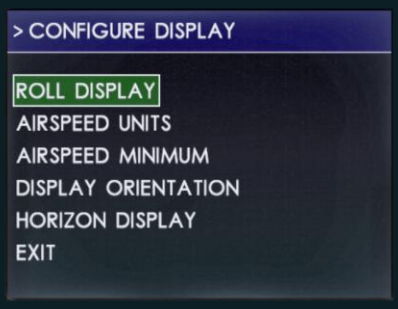
PT 7: photocell at or below this input

DIMMING CURVE with INTERNAL: input from MIN through AMBIENT to MAX light, backlight from LO to HI; DIMMING CURVE with EXT 28V: input from 0 to 28 V. The vertical line shows where the knob is now; DIMMING CURVE, PT 7: the input at or below which the photocell takes over

For what the default curve does as the panel-light knob is turned, see *How the automatic dimming works*.

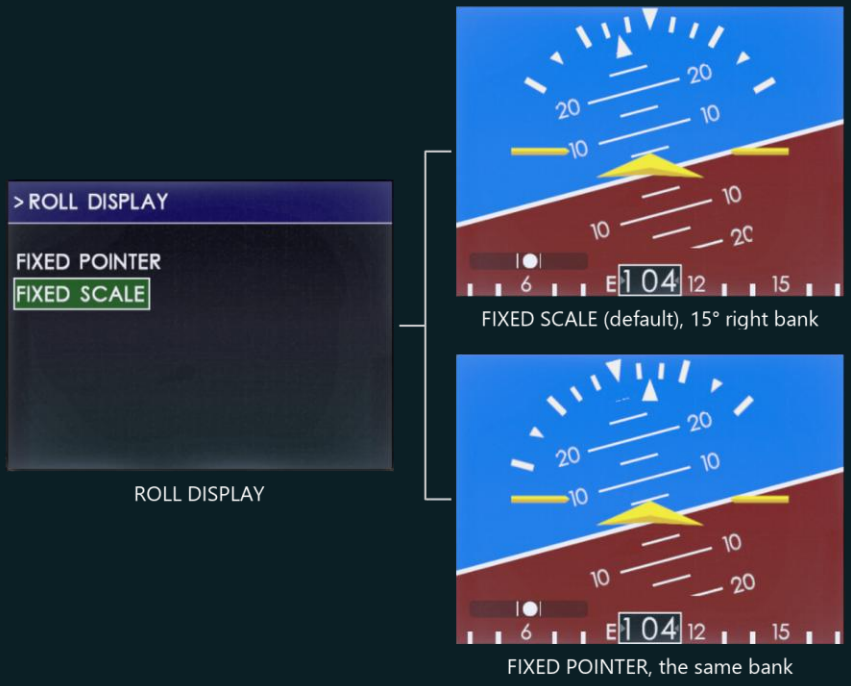
Configure Display

There are five display choices. On the ground the attitude is level, so the preview does not show the difference in ROLL DISPLAY. The pictures below are in a 15° right bank.



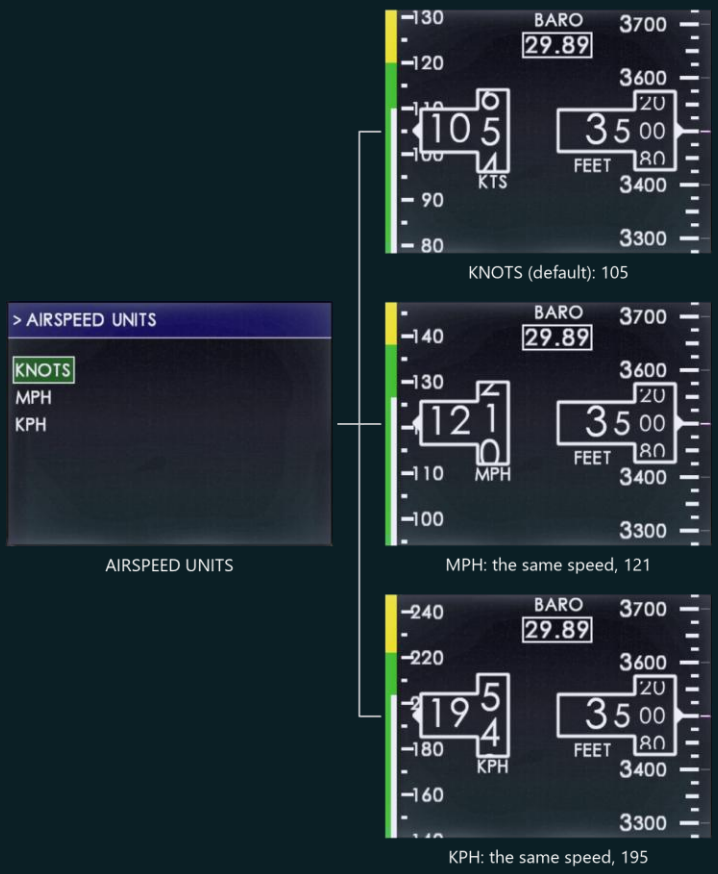
CONFIGURE DISPLAY: ROLL DISPLAY, AIRSPEED UNITS, AIRSPEED MINIMUM, DISPLAY ORIENTATION, HORIZON DISPLAY

ROLL DISPLAY: which part moves with the bank. With FIXED SCALE, the scale stays at the top and the small triangle pointer turns with the horizon. With FIXED POINTER, the scale turns with the horizon and the pointer stays at the top.



ROLL DISPLAY: FIXED POINTER or FIXED SCALE

AIRSPEED UNITS: KNOTS, MPH or KPH. The tape is converted, and so are the bands that come from the aircraft's V-speeds. Lines typed by hand and the VNE TABLE keep their numbers, so they are read in the new units. KTS under the window becomes MPH or KPH. For example, about 105 kt reads about 121 mph or 195 km/h. While it is being set, the preview changes to the airspeed display.



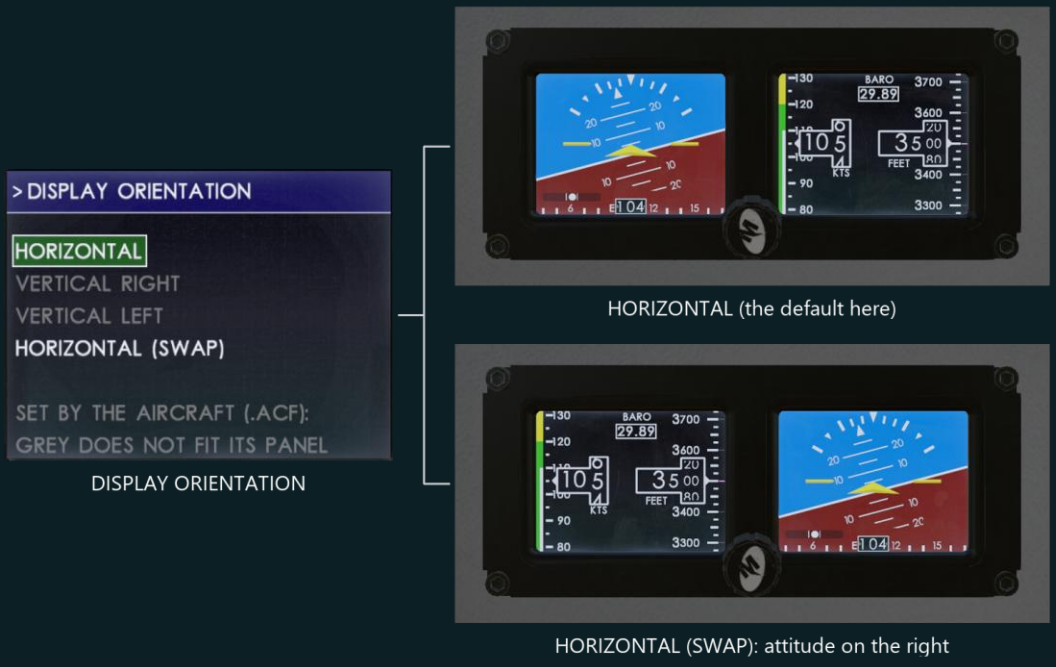
AIRSPEED UNITS: KNOTS, MPH or KPH

AIRSPEED MINIMUM: 20, 30 or 40 KNOTS. Below it the airspeed reads 0, so it does not jump about while taxiing. At 25 kt, a minimum of 20 shows 25, while 30 or 40 show 0. At 35 kt, 20 or 30 show 35, while 40 shows 0.



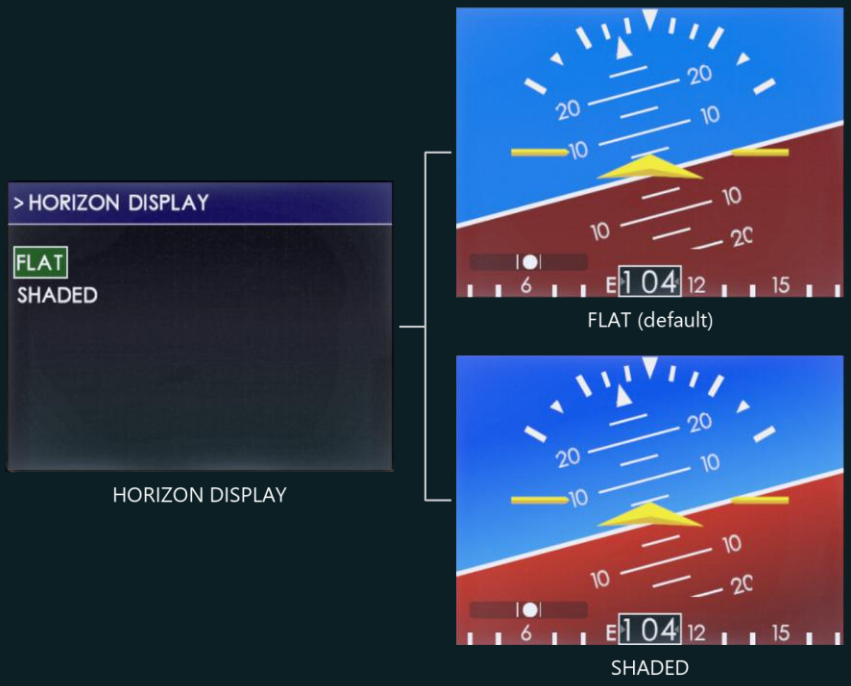
AIRSPEED MINIMUM: 20, 30 or 40 KNOTS

DISPLAY ORIENTATION: which display goes where. HORIZONTAL (SWAP) puts the attitude display on the right and the airspeed and altitude display on the left. Choices that do not fit how the unit is mounted in the aircraft are grey. Here those are VERTICAL RIGHT and VERTICAL LEFT, which need a vertical installation (see *Vertical installation*).



DISPLAY ORIENTATION: only HORIZONTAL and HORIZONTAL (SWAP) fit this panel; DISPLAY ORIENTATION HORIZONTAL (the default here, where auto follows the .acf); DISPLAY ORIENTATION HORIZONTAL (SWAP): attitude on the right

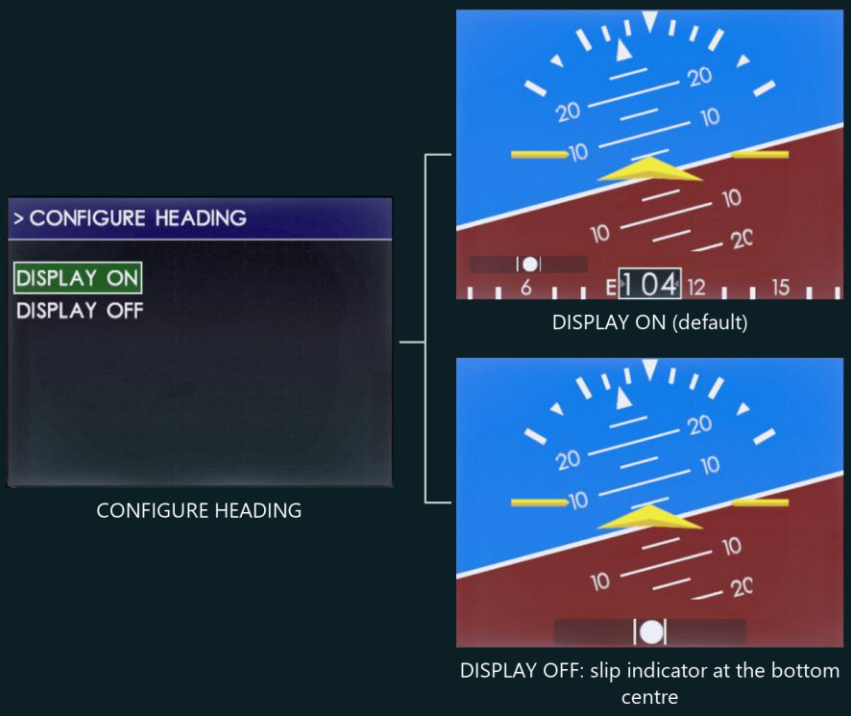
HORIZON DISPLAY: FLAT fills the sky and the ground with one even blue and one even red. SHADED grades the colors. The sky, for example, is lighter near the horizon and darker toward the top.



HORIZON DISPLAY: FLAT or SHADED

Configure Heading

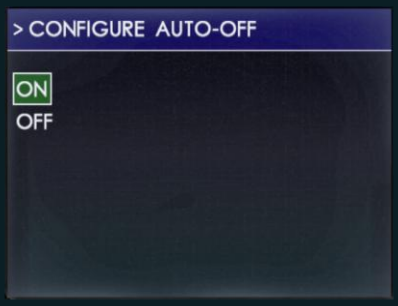
DISPLAY OFF removes the heading window and tape from the bottom of the attitude display, and the slip indicator moves to the bottom center.



CONFIGURE HEADING: DISPLAY ON or DISPLAY OFF

Configure Auto-Off

AUTO-OFF decides whether the unit counts down to turning itself off when aircraft power is lost on the ground.



CONFIGURE AUTO-OFF: ON or OFF

With ON (the default), the POWER DOWN IN 60 SEC box comes up, as shown under "*On the ground*". With OFF there is no box. The unit runs on its battery, with the green battery icon at the top right, until it is turned off with POWER OFF in the Options Menu.

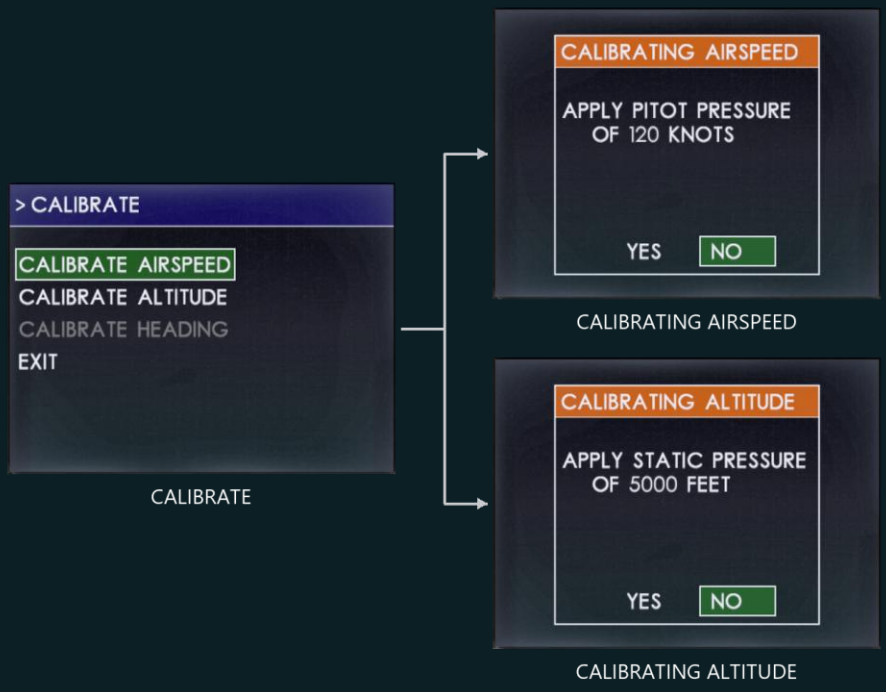


AUTO-OFF OFF: aircraft power off on the ground, no countdown, green battery icon

Calibrate and Update Software

These screens look like the real unit's, but they change nothing, because the simulator has no test equipment and no USB port.

CALIBRATE AIRSPEED asks for 120 kt of pitot pressure, and CALIBRATE ALTITUDE for 5,000 ft of static pressure. Whether you answer YES or NO, no value changes.



CALIBRATE: CALIBRATE AIRSPEED and CALIBRATE ALTITUDE, with CALIBRATE HEADING grey; CALIBRATING AIRSPEED: APPLY PITOT PRESSURE OF 120 KNOTS; CALIBRATING ALTITUDE: APPLY STATIC PRESSURE OF 5000 FEET

UPDATE SOFTWARE **always ends in FAILED UPDATE**; OK goes back to the menu. It is grey while the unit runs on its battery.

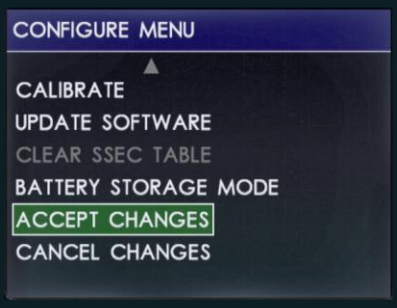


UPDATE SOFTWARE: FAILED UPDATE

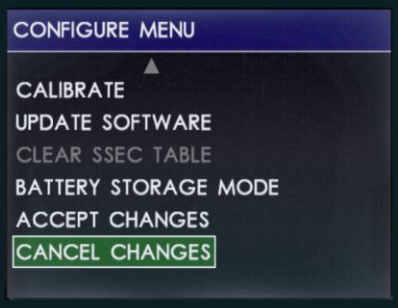
Saving

Nothing is saved until ACCEPT CHANGES. Every setting is a line in `config.ini`. The settings only the aircraft's author changes, such as the heading source, are described under *Installation and Settings*.

ACCEPT CHANGES saves every change, and CANCEL CHANGES drops them all. Either way the unit restarts and shows the start-up screen for about 5 s.



ACCEPT CHANGES: save and restart



CANCEL CHANGES: drop and restart

Cursor on ACCEPT CHANGES: save every change and restart; Cursor on CANCEL CHANGES: drop every change and restart

Installation and Settings (for Aircraft Authors)

See also: rev Q section 3.

This chapter is for the author who puts the MD302 in an aircraft. It covers what the unit needs, how to install it, and the settings only the author changes. Pilots can skip it.

Requirements

- X-Plane 12.4.4 with Panel Graphics (tested on 12.4.4-b5). Tested on Windows & Mac only, Linux untested.
- xLua 2.0 in the aircraft's *plugins/xlua/* folder, tested with 2.0.0-b5. Earlier builds can run out of callbacks after the pop-up has been opened many times. xLua 2 also runs the aircraft's existing xLua 1 scripts.

Installing

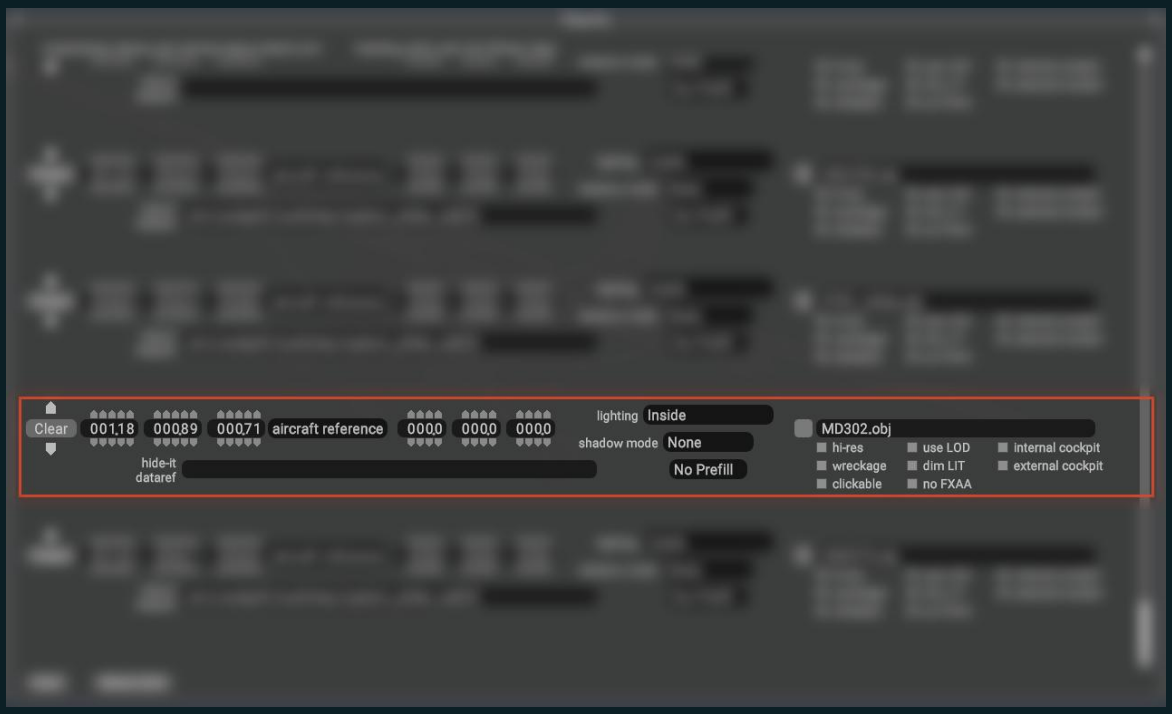
1. Copy the MD302 folder to *<aircraft>/plugins/xlua/scripts/MD302/*. The folder **must** be named *MD302*.
2. In Plane Maker, attach *../plugins/xlua/scripts/MD302/3D/MD302.obj* as a misc object. It is an ordinary object, not the cockpit object. Its origin is the center of the back of the bezel, so put it on the panel surface. The front faces the pilot, so an upright panel needs no rotation.
3. Set *power_bus* in *config.ini* to the electrical bus that feeds the unit (below).
4. Check *air_data*: the default, *stby*, is X-Plane's standby pitot-static system. Some aircraft plugins take over the indicated airspeed and feed only the pilot's side (c172_IAS.lua in X-Plane's Cessna 172 does). The standby airspeed then stays at 0, so the unit reads X-Plane's standby calibrated airspeed instead. If it still reads 0 in flight, set *pilot* or *copilot*.
5. For a vertical installation, roll the object a quarter turn in Plane Maker. A roll of -90 puts the knob on the right, and +90 on the left. PANEL TILT and PANEL ROLL take out a small tilt or roll of the object.
6. Choose the heading source, *heading* (below). As shipped it shows X-Plane's heading.

Placing the unit by eye *(sorry, there's really no better way around)*

The origin of *MD302.obj* is the geometric center of the back of the bezel, where the bezel meets the panel. Put that point in the middle of the panel hole, on the panel surface, and the unit sits flush.

Part	Size and place (from ***MD302.obj***)
Axes	x to the right, y up; the front faces aft, toward the pilot
Bezel (front plate)	139.7 x 60.2 mm (5.50 x 2.37 in), 8.4 mm thick, in front of the panel
Case	about 133 x 58 mm, 120.5 mm deep, behind the panel
Knob	center 20.8 mm below the middle of the bezel; its front 24.9 mm in front of the panel

In Plane Maker, open Standard > Objects. Each object is one row; the MD302 row of our test aircraft is outlined below.



Plane Maker Standard > Objects, lighting = Inside, shadow mode = None (or your preference)

Set **lighting** to inside. The test aircraft uses Inside and shadow mode None, but feel free to configurate this according to your own desired art taste.

The settings file

config.ini in the MD302 folder holds every setting, one *key = value* per line. *;* or *#* at the start of a line or after a space starts a comment. Each line is explained in the file itself.

Part of the file	Written by
[Options]	The Options Menu, the moment a value is picked
[Configure ...] (all but *dimming_rheostat*)	Configuration Mode, on ACCEPT CHANGES
[State]	The unit: battery charge and total run time, kept from one flight to the next
[Power], *[Sources]*, *[Display]*, *[Clicks]*, *dimming_rheostat*	Only the aircraft's author, by hand. (ACCEPT CHANGES writes *baro_sync_source* back, unchanged, when CONFIGURE BARO SYNC was changed.)

- The unit rewrites single lines and keeps the rest, comments included. A save goes to *config.ini.tmp* first, and *config.ini.bak* keeps the version before the last save. If X-Plane stops half-way, or the file is found empty or damaged (unreadable, holding NUL bytes, or with no setting, comment or section line), the next load puts it back from the *.tmp* or the *.bak*. The damaged file is kept as *config.ini.broken*.
- If *config.ini* is deleted, an existing *config.ini.bak* is put back. With neither, the next save writes a new *config.ini* with only the values it saved, without the explanations. To start again from the factory settings, copy a fresh *config.ini* from the download and delete *config.ini.bak*.

- A value that is not allowed is replaced and reported in Log.txt. A number with limits takes the nearest limit. A *panel_tilt* or *panel_roll* out of range becomes *auto*, an *airspeed_minimum* other than 20, 30 or 40 becomes 30, a bad dimming curve becomes the whole default curve, and a bad *vne_table* is emptied. Anything else becomes its default. The SETTINGS MISSING OR INVALID box also comes up when the unit first reaches flight mode. Two cases give no box, only a line in Log.txt: a value that is not a number, and a bad *rangeN* line. An unknown key is ignored and reported in Log.txt.

Power and battery

Setting	Default	What it does
power_bus	0	The electrical bus that feeds the unit: the index (0–5) of X-Plane's *sim/cockpit2/electrical/bus_volts*
power_min_volts	10	Below this the unit counts as unpowered (the real unit runs on 10–32 V)
battery_type	1	*1* = lithium-ion. *2* = nickel metal hydride, about 145 minutes when new (set *battery_minutes* to match). *none* = no battery. The unit goes off as soon as aircraft power goes, with no battery icon and no battery check at start-up. BATTERY INFO and BATTERY STORAGE MODE are grey
battery_minutes	120	Minutes of operation on a full battery (rev Q: at least one hour, up to 2.5 hours when new)
battery_charge_minutes	180	Minutes to recharge from empty
battery_low_pct	20	Below this charge the battery icon turns black with a red X
preflight_s	5	Seconds the start-up screen shows. It is also the way into Configuration Mode, so keep it a few seconds long
power_down_countdown_s	60	Seconds the power-down countdown runs on the ground (AUTO-OFF)
ground_ias_kt	20	Airspeed in knots below which the unit counts as on the ground. AUTO-OFF uses only this, so taxiing counts too; Configuration Mode needs it as well
ground_speed_kt	5	Ground speed in knots that must also be below this value for Configuration Mode. AUTO-OFF ignores it

Data sources

Setting	Default	Choices
air_data	*stby*	*stby* = X-Plane's standby pitot-static system and its own baro setting, like the real unit's own ports. Standby pitot and static failures apply. When an aircraft plugin takes over the indicated airspeed and leaves the side read at 0, the unit reads that side's calibrated airspeed instead. *pilot* or *copilot* = that side's system. Failures of that side's airspeed indicator and altimeter give the red X
attitude	*own*	*own* = the aircraft's attitude, standing in for the MD302's own sensors. It never drifts. *ahrs*, *electric* or *vacuum* = that pilot-side X-Plane instrument, with its errors and failures
baro_sync_source	*pilot*	Whose altimeter setting CONFIGURE BARO SYNC follows when it is on (*baro_sync*, below): *pilot* or *copilot*
heading	*pretend*	What feeds the heading window (see the next table)

The *heading* choices:

heading	What the heading window shows
pretend (as shipped)	X-Plane's magnetic heading whenever the unit is on, with the ARINC arrows. It is never lost. A convenience for the simulator, not a real MD302 setting
none	"---", as a real unit with no heading source wired
md32	An MD32 magnetometer, giving X-Plane's magnetic heading with the arrows pointing out. It runs on the MD302's own power and is never lost
arinc	The pilot's AHRS heading, with the arrows pointing in. It is lost while the aircraft has no power or X-Plane's AHRS is failed. The unit then runs 10 s on its own gyro without the arrows, then shows "---". The heading comes back with the source

Dimming input

dimming_rheostat (default 0) picks the aircraft's panel-light knob that EXT 5V, EXT 14V and EXT 28V dimming follow. The value is the index (0–31) of X-Plane's *sim/cockpit2/electrical/instrument_brightness_ratio_manual*. This input reads 0 while aircraft power is lost.

Configuration Mode settings in the file

The *[Options]* lines (*alt_units*, *baro_units*, *symbol*, *att_mask*, *alt_trend*) are the Options Menu items. The *[Configure ...]* lines below are what Configuration Mode sets. Set them by hand to ship the aircraft already configured. The range markings, the VNE table and the SSEC name are in the next section.

Setting	Configuration Mode item	Default	Other values
panel_tilt	PANEL TILT	*auto* (from how the object sits in Plane Maker)	-90 to 90
panel_roll	PANEL ROLL	*auto*	-5 to 5, in 0.5 steps

Setting	Configuration Mode item	Default	Other values
range_mmo	RANGE MMO	0 (no barber pole)	up to 0.999
baro_sync	CONFIGURE BARO SYNC	*off*	*on*: follows *baro_sync_source*
arinc_pass_through	PASS-THROUGH	*off*	*on*
arinc_speed	TRANSMIT SPEED	*high*	*low*
dimming_control	DIMMING CONTROL	*internal*	*ext_5v*, *ext_14v*, *ext_28v*
dimming_curve_internal, *dimming_curve_external*	DIMMING CURVE	*10 28 46 64 82 100*	six levels, each 5 to 100 %, at 0, 20, 40, 60, 80 and 100 % input
dimming_low_input	DIMMING CURVE, PT 7 (ext. dimming only)	0	0 to 100 % of the input. At or below this the photocell takes over
roll_display	ROLL DISPLAY	*fixed_scale*	*fixed_pointer*
airspeed_units	AIRSPEED UNITS	*knots*	*mph*, *kph*
airspeed_minimum	AIRSPEED MINIMUM	30	20, 40 (always knots; below it the airspeed reads 0)
display_orientation	DISPLAY ORIENTATION	*auto* (from Plane Maker)	*horizontal_swap* (attitude on the right LCD; horizontal installation only). *horizontal*, *vertical_right* and *vertical_left* count only when the .acf has no MD302.obj row.
horizon_display	HORIZON DISPLAY	*flat*	*shaded*
heading_display	CONFIGURE HEADING	*on*	*off*
auto_off	CONFIGURE AUTO-OFF	*on*	*off*

Airspeed markings and tables in the file

Configuration Mode writes *range_markings* (with *range1* to *range8*) and *vne_table*; setting them by hand gives the same result. *ssec_file* is only set by hand.

- *range_markings = auto* (default) takes the bands from the aircraft's V-speeds in Plane Maker: full white Vso–Vs, half white Vso–Vfe, full green Vs–Vno, full yellow Vno–Vne, full red from Vne up.
- *range_markings = manual* uses up to eight lines, *range1* to *range8*. Each is *TYPE CLR V1 V2*, as on the RANGE MARKINGS page. TYPE is *FULL*, *HALF*, *RAD* or *OFF*. CLR is *WHT*, *GRN*, *YEL*, *RED*, *BLU* or *BAR* (barber pole). Speeds are in the AIRSPEED UNITS, and a V2 of *MAX* (or 999) runs to the top of the tape. Example: *FULL WHT 62 68*.
- *vne_table*: up to 7 points, *feet speed ...*. The first is at 0 ft, the altitudes never go down (two points at one altitude make a step), and the speeds are in the AIRSPEED UNITS. Empty means no table. Example: *0 250 12000 250 17000 230*.
- *ssec_file*: the name of an SSEC table. The start-up screen shows it as SSEC: FILE followed by the name, with the name in capitals and without spaces. No correction is applied. Empty means none. It is set only here, since Configuration Mode can only clear it (CLEAR SSEC TABLE).

Display and pop-up

Setting	Default	What it does
enable_popup	1	0 = no pop-up window at all
popup_initial	0	1 = the pop-up opens when the aircraft loads
lcd_effects	1	1 = the LCD look (band, glow, grain) is drawn over the picture; 0 = a clean picture
lcd_band	1	The backlight showing through the black, 0–1. 1 = as on the real unit in the 3D cockpit; 0 = pure black. X-Plane's panel light does not brighten the pop-up, so there the band looks nearly black
lcd_bleed	0.15	The glow round the edges, 0–1
lcd_grain	0.1	The pixel-to-pixel unevenness, 0–1
screen_fb, *left_lcd_x*, *left_lcd_y*, *right_lcd_x*, *right_lcd_y*	—	The size of the drawing surface and where the displays sit on it, matching the model's texture. Leave as shipped

Clicks in the 3D cockpit

X-Plane handles clicks only on the aircraft's cockpit object, and *MD302.obj* is an ordinary object. So the MD302 picks the clicks on its knob and screens itself. This works in the 3D cockpit view, not in VR or outside views.

Setting	Default	What it does
pick_mode	1	0 = off; 1 = auto (off when X-Plane can already click the unit); 2 = always, unless *MD302.obj* is itself the cockpit object
pick_pass_misses	0	A click inside the MD302's click area that misses the unit: 0 = the MD302 takes it, 1 = X-Plane gets it
pick_debug	0	1 = write the click areas and the hits to Log.txt

When X-Plane's screen and its 3D view differ in shape by more than 2 %, the 3D clicks are turned off and Log.txt says so once. This happens with some multi-monitor, windowed or high-DPI layouts. The pop-up and the commands still work.

Commands

Each command can be assigned to a key, a joystick or a VR controller button in X-Plane's settings, under the name in the second column.

Command	Name in X-Plane	What it does
laminar/md302/press_knob	MD302 knob push (hold 2 s: Options Menu)	Pushes the knob for as long as it is held
laminar/md302/knob_left	MD302 knob turn left	One detent counter-clockwise
laminar/md302/knob_right	MD302 knob turn right	One detent clockwise

Command	Name in X-Plane	What it does
laminar/md302/goes_into_setting	MD302 open the Options Menu	Opens the Options Menu at once, without holding (in flight mode, with no box up)
laminar/md302/power_down_off	MD302 power-down: remain on (ON)	Answers the power-down countdown with ON (also in advance), so the unit stays on its battery. Does nothing on aircraft power
laminar/md302/popup_toggle	MD302 toggle popup window	Opens or closes the pop-up
laminar/md302/popup_show	MD302 show popup window	Opens the pop-up, as a click on a 3D screen does
laminar/md302/config_mode	MD302 restart in Configuration Mode (on the ground)	Opens Configuration Mode, as holding the knob at power-up does. Works only with the unit on, on the ground and on aircraft power

goes_into_setting and *power_down_off* keep their names from the 2024 version of the MD302.

Datarefs

The four writable datarefs, for an instructor or a scenario, are listed under *Failures for instructors*. All the others are read only. Use them for a warning light, an animation or a panel of the aircraft's own.

Dataref (*laminar/md302/...*)	Value
valid	The real unit's valid output, for a warning light. 1 in flight mode with no red X; 0 when off, starting, in Configuration Mode or with a red X
unit_on	1 = the unit is on
ops_mode_now	-1 off, 0 start-up, 1 flight, 3 Options Menu or Configuration Mode
on_battery	1 = running on its battery
battery_pct	Battery charge, %
power_down_timer_remain	Seconds left in the power-down countdown
menu_open	1 = the Options Menu is open
config_mode	1 = in Configuration Mode
alt_trend_fpm	The vertical speed behind the altitude trend bar, ft/min
knob_val_for_setting	Knob rotation for the 3D model: 1 = one turn (16 detents)
knob_push_anim	Knob push for the 3D model: 0 out to 1 pushed in
knob_pressed	1 = the knob is held in
popup_visible	1 = the pop-up is open
xlua_heartbeat	Counts up while the script runs
cursor_report	Text, for diagnostics: which part of the unit X-Plane last asked about the mouse, and where the pop-up is. Missing if X-Plane offers no byte-data datarefs

Kept from the 2024 version (Eric's work in the pre-AI era), so panels made for it keep working:

Dataref (*laminar/md302/...*)	**Value**
options_alt_units_now	0 feet, 1 metres
options_baro_units_now	0 hPa, 1 inHg
options_symbol_now	0 DELTA, 1 TRADITIONAL, 2 SPLIT DELTA
options_att_mask_now	1 = ATT MASK on
options_alt_trend_now	1 = ALT TREND on
on_ground	1 = on the ground: ground speed below *ground_speed_kt* and airspeed below *ground_ias_kt*
pressure_display_val	The BARO value shown, in hPa or inHg

Not Simulated in X-Plane

See also: rev Q sections 4.5.9, 4.5.10, 4.5.11, 5.1.1.

Some parts are shown and can be stepped through, but change nothing in the simulator; others are not reproduced at all. The full list:

- CALIBRATE: nothing is calibrated or saved; CALIBRATE HEADING is grey.
- ARINC 429 settings (PASS-THROUGH, TRANSMIT SPEED): stored and shown only.
- SSEC: a table can be named, but no correction is applied.
- UPDATE SOFTWARE: always ends in FAILED UPDATE (grey on the battery).
- The real unit's AHRS drift and errors: the attitude is X-Plane's. When the author connects the attitude to the AHRS, electric or vacuum gyro, that instrument's X-Plane errors come with it.
- Battery heating and ageing, and the cold-weather behaviour of the displays. BATTERY INFO's temperature is the outside air, held between 5 and 40 °C.
- The default heading never fails, as a convenience. A real unit needs a heading source.
- The knob works in the 3D cockpit view, not in VR or outside views: use the commands instead.
- The power-up self-test never fails on its own. Apart from the battery and settings checks, the message boxes come up only from an instructor's `fail_message`.
- Each display is drawn as a 500 x 374 picture, not 320 x 240 pixels. The pixel grid and the off-axis glow of the real screen are approximated, not reproduced.

Tutorial: Losing Aircraft Power and Real Failures

See also: rev Q sections 4.4.1, 4.4.2.

The MD302 carries its own battery, so losing aircraft power is not a failure of the MD302. It goes on running on its battery (unless it is a version without one). It fails only when the battery runs out, or when one of its inputs fails.

Part 1: Aircraft power lost in flight

- In flight, turn the aircraft's battery and alternator off (or fail the electrical bus in Flight > Failures).
- The GPS, the autopilot and most of the radio stack go dark. Instruments with batteries of their own (in our test aircraft, a C172, the G5s) and the transponder stay on. The MD302 keeps running, and the green battery icon appears at the top right of its attitude display.
- There is nothing to acknowledge. Keep flying on the MD302's attitude, airspeed and altitude.



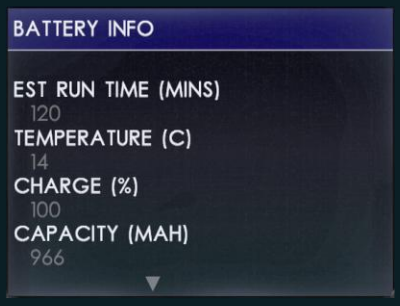
Aircraft power off in flight: the MD302 runs on its battery (the rest of the panel is blurred)



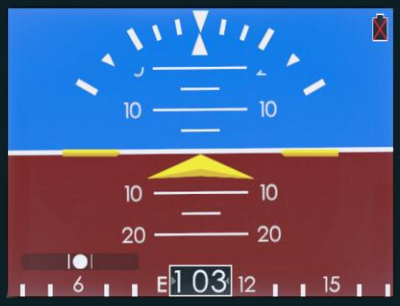
The MD302 on its battery: the green battery icon at the top right

Part 2: The battery running low

- In X-Plane a full battery lasts 120 minutes by default. The real unit gives at least one hour, and up to about 2.5 hours when new. BATTERY INFO in the Options Menu shows the run time left.
- Below 20 % the icon turns black with a red X. Land soon.
- When the battery is empty, the MD302 turns off and both displays go black, with no warning box. With the default battery, that is after 120 minutes on a full charge, about 24 minutes after the red X. When aircraft power is back, the MD302 starts again and recharges (3 hours from empty). At that start-up it warns that the battery may last less than 60 minutes.



BATTERY INFO, full battery: 120 min

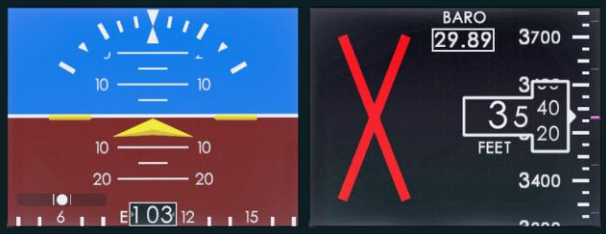


Below 20 %: black icon, red X

BATTERY INFO on a full battery: 120 min of run time; Below 20 %: the black battery icon with a red X

Part 3: A real failure on the input side

- Fail an instrument the MD302 reads (for example the pilot's airspeed indicator, when the aircraft's author connected it to the pilot side), or set `laminar/md302/fail_airspeed` to 1.
- A red X covers only that part, and attitude and altitude keep working.
- A blocked pitot tube or static port gives no red X: the MD302 shows the wrong values, as a real one would.



Only the airspeed failed: a red X over the airspeed tape. Attitude and altitude keep working

Part 4: After landing

- If the unit is still on the battery with no airspeed (below 20 kt, so taxiing counts too) and AUTO-OFF is on, POWER DOWN IN 60 SEC counts down.
- Select ON to keep it running on the battery. OFF is lit first, so a push turns it off, and so does the count reaching 0. An airspeed of 20 kt or more takes the box away; if the airspeed drops again, the count starts again from 60.
- Turn aircraft power back on. If the MD302 had turned off, it starts again on aircraft power; if it was still running, it goes back to aircraft power. Either way it recharges its battery.



The power-down countdown after landing

References

Mid-Continent Instruments and Avionics, *MD302 Series Standby Attitude Module (SAM) Installation Manual and Operating Instructions*, Manual Number 9017782, Revision Q, March 12, 2024. Section numbers in this manual refer to the Mid-Continent manual above.

Appendix A

General description (real manual section 1)

Function	Real unit	X-Plane MD302	Status
Displays	Two 2.4-inch color LCDs of 320 x 240 pixels.	Two screens drawn live in the cockpit with the look of the real LCDs (backlight glow, brighter edges, slight pixel grain). The pixel grid is imitated, not reproduced.	**Partial**
Orientation	“AnyWay” installation: horizontal, or vertical with the knob on the left or right.	Horizontal, horizontal with the screens swapped, and vertical either way. The unit reads how it sits in Plane Maker and turns its pictures to match.	**Same**
Aircraft power	Runs on 10 to 32 V DC aircraft power.	Runs from the electrical bus chosen by the aircraft’s author; below 10 V it counts as unpowered.	**Same**
Internal battery	Rechargeable lithium-ion or nickel metal hydride battery with a heater; at least one hour of backup (up to about 2.5 hours when new).	Drains in use and recharges on aircraft power: 120 minutes by default, lithium-ion or nickel metal hydride, or a version with no battery. No temperature, heater or ageing.	**Partial**
Lighting	Follows the aircraft’s 5, 14 or 28 V lighting, or its own photocell.	Photocell: follows the daylight in X-Plane. External: follows the aircraft’s panel-light knob.	**Same**
Attitude sensing	Its own solid-state attitude sensors.	Shows X-Plane’s aircraft attitude, so it never drifts. The author can choose to show the aircraft’s AHRS or gyro instead, with their errors and failures.	**Partial**
Air data	Its own airspeed and altitude sensors, connected to the aircraft’s pitot and static lines.	Reads X-Plane’s standby pitot-static system by default (or the pilot’s or co-pilot’s). A blocked pitot or static port shows what it would on the real unit.	**Same**
Operating limits	Pitch and roll rates up to 300° per second; altitude -1,500 to 55,000 ft; airspeed 20 to 500 kt.	Same limits. Beyond them a red X covers the attitude (until 3 s after the rate is back in range) or the altimeter. The airspeed shows a red X above 500 kt, and reads 0 below the airspeed minimum.	**Same**
ARINC 429 data	Receives the baro setting and heading; sends attitude, altitude and airspeed to other avionics.	Baro sync and heading come from X-Plane’s own instruments. Nothing is sent out.	**Partial**
Valid output	An electrical “valid” signal, for a warning light.	A dataref the aircraft’s author can use for a warning light.	**Same**

Installation (real manual section 3)

Function	Real unit	X-Plane MD302	Status
Installation	Cable harness with configuration module, pitot-static connections, panel mounting.	Placed as an object in Plane Maker; the power source and the data sources are set in the unit’s settings file.	**Partial**

User interface and pre-flight mode (real manual sections 4.1 and 4.2)

Function	Real unit	X-Plane MD302	Status
Control knob	One push-and-turn knob, 16 detents per turn.	Same. With the mouse: click the left or right half to turn it, the middle to push it, or use the scroll wheel.	**Same**
Start-up screen	About five seconds: logo, model, SSEC file, software version, total operating time.	Same. The software version reads “X-PLANE 2.0.0”.	**Same**
Power-up self-test	Checks the battery, the software and memory, and the configuration module; reports any fault.	Checks the battery and the settings file. Other faults appear only when an instructor sets them.	**Partial**

Flight mode (real manual section 4.3)

Function	Real unit	X-Plane MD302	Status
Attitude	Roll scale (fixed scale or fixed pointer), pitch ladder through 360°, red chevrons at extreme pitch.	Same.	**Same**
Heading	Heading window and moving tape, from an MD32 magnetometer or an ARINC 429 source; “---” with no source; 10 seconds on its own gyro when the source is lost.	Same, with the source chosen in the settings file. As shipped it shows X-Plane’s heading at all times (see Added for X-Plane). The MD32 never fails.	**Partial**
Slip	Slip indicator.	Same.	**Same**
Altitude	Altitude tape in feet or meters, baro setting in inHg or hPa (28.00 to 31.00 inHg), 6-second altitude trend.	Same.	**Same**
Airspeed	Airspeed tape in knots, mph or km/h; colored range markings and barber pole; reads 0 below the airspeed minimum.	Same. If the installer has set no markings, they are taken from the aircraft’s speeds in Plane Maker.	**Same**
Options Menu	ALT UNITS, BARO UNITS, SYMBOL, ATT MASK, ALT TREND, INFO (REVIEW CONFIG, SOFTWARE INFO, BATTERY INFO, EXIT INFO), EXIT MENU, POWER OFF.	Same. Every choice is remembered after a reload.	**Same**
Battery information	Run time left, temperature, charge, capacity and type.	Same page. The temperature follows the outside air (held to 5 to 40 °C) and the capacity never falls.	**Partial**
Brightness	Push the knob, then turn it; cannot go below the dimming curve.	Same.	**Same**

Emergency operation (real manual section 4.4)

Function	Real unit	X-Plane MD302	Status
Power lost in flight	The battery takes over without interruption: green battery icon; black icon with a red X below 20 %.	Same.	**Same**
Power lost on the ground	With no airspeed, a 60-second POWER DOWN countdown, answered ON or OFF; Auto-Off can be turned off.	Same.	**Same**
POWER OFF	Turns the unit off while it runs on the battery.	Same, in the air too (service bulletin SB MD302-09).	**Same**

Configuration Mode (real manual section 4.5)

Function	Real unit	X-Plane MD302	Status
Entering	Hold the knob while aircraft power is applied, on the ground.	Same, plus two easier ways: hold the knob for 3.5 seconds during the start-up screen, or use a command. On the ground and on aircraft power only.	**Partial**
Configure preview	A preview of the setting being changed.	Same: the attitude screen shows the setting live.	**Same**
Configure Aircraft	Panel tilt, panel roll, Mmo, range markings, Vne table.	Same. Panel tilt and roll can be left on automatic, taken from how the unit sits in the cockpit.	**Same**
Configure ARINC	Pass-through and transmit speed.	Stored and shown only.	**Partial**
Configure Baro Sync	Takes the baro setting from the main avionics.	Follows the pilot’s or co-pilot’s altimeter in X-Plane.	**Same**
Configure Dimming	Internal or external (5, 14 or 28 V) control, and the dimming curve.	Same.	**Same**
Configure Display	Roll display, airspeed units, airspeed minimum, display orientation, flat or shaded horizon.	Same, but the orientation must match how the unit sits in Plane Maker: the other choices are grey.	**Partial**
Configure Heading	Heading display on or off.	Same.	**Same**
Configure Auto-Off	Auto-Off on or off.	Same.	**Same**
Calibrate airspeed and altitude	Corrects sensor offsets with a pitot-static tester.	The screens are there, but nothing is calibrated or saved.	**Partial**
Calibrate heading	Calibrates the MD32 magnetometer in the aircraft.	Grey: not available.	**Not modelled**
Update software	Loads new software from a USB drive.	There is no USB port: it always ends with “FAILED UPDATE”.	**Not modelled**
SSEC table	Static source error correction table.	A table can be named and cleared, but no correction is applied.	**Partial**

Function	Real unit	X-Plane MD302	Status
Battery storage mode	Turns the battery off for storage (service bulletin SB MD302-05).	Same: the battery stays off until aircraft power comes back.	**Same**
Accept or cancel changes	Nothing is saved until ACCEPT CHANGES.	Same: accepted settings are written to the settings file and the unit restarts.	**Same**

Continued airworthiness and limitations (real manual sections 5.1 and 2.2)

Function	Real unit	X-Plane MD302	Status
Warning and error messages	Battery capacity warning and troubleshooting messages at power-up.	Same wording and colors. The battery warning and the settings-file messages come up by themselves; the others only when an instructor sets them.	**Partial**
Battery over temperature	Shown when the battery is too hot.	Only when an instructor sets it.	**Partial**
Maintenance	Battery replacement, pressure-system checks, software updates.	Not applicable in the simulator.	**Not modelled**
Cold weather	Below -30 °C the screen can take up to five minutes to become fully readable.	Not modelled.	**Not modelled**